

Tail-Index Sure Independence Screening for Ultra-High-Dimensional Heavy-Tailed Regression

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Abstract

We propose a marginal screen for the covariates that determine a positive conditional tail index, designed for problems where the dimension p is large relative to the number of observations informative about the tail. The screen rests on one geometric fact: after projection onto a single coordinate, the conditional distribution is a mixture whose tail is governed by its heaviest component, so the projected tail index is the upper envelope of the tail-index surface along the corresponding fibre. Inactive coordinates produce flat envelopes at the global maximum γ^* ; active coordinates are detectable exactly when their fibres miss the global argmax set on a set of positive measure. Ranking coordinates by the average projected index therefore separates detectable active variables from noise. We estimate the envelope with empirical-rank local Hill profiles, and prove that the maximal score error over p coordinates is $O_{\mathbb{P}}(\sqrt{\log(pn)/(nah)})$ plus a bias term negligible relative to the population gap, where α is the local tail fraction and h the rank bandwidth; nah is the effective local extreme sample size. Sure screening and exact recovery follow when $\log(pn) = o(nah\Delta_{\min}^2)$. The algorithm costs $O(pn \log n)$. Simulations with slowly varying factors driven by inactive covariates show when screening on the tail index differs from screening on a fixed high quantile. An application to violent-crime rates in 1,993 U.S. communities, with $p = 99$ continuous predictors against an effective tail sample of about 200, illustrates the method in its intended regime. The uniform projected-quantile conditions are stated directly; deriving them from a fully primitive growing-dimensional model remains open.

1 Introduction

Let (X, Y) be a random pair with $Y > 0$ and $X \in \mathbb{R}^p$, and suppose that, conditionally on $X = x$,

$$\frac{\mathbb{P}(Y > ty \mid X = x)}{\mathbb{P}(Y > y \mid X = x)} \longrightarrow t^{-1/\gamma(x)}, \quad y \rightarrow \infty, \quad t > 0, \quad (1)$$

for a positive function γ , the conditional tail index. Estimating γ is the first step toward extreme conditional quantiles and tail probabilities [6, 12], and it suffers from a particularly severe curse of dimensionality: local estimation must find observations near x and then keep only an upper fraction of their responses. A one-dimensional window of width h combined with a tail fraction α leaves an effective sample of order nah , not n [5, 10]. When p is large, even moderate sample sizes are effectively small.

This paper asks the screening question: among p candidate covariates, possibly with p growing with n , which ones does γ actually depend on? Sure independence screening [7] answers such questions by ranking coordinates through inexpensive marginal statistics and keeping a moderate number for subsequent joint analysis. Model-free variants exist for general dependence [13] and for fixed conditional quantiles [11], but the limiting exponent in (1) is a different target: a covariate can move the 0.95 quantile of Y without affecting γ , and conversely. Our simulations make this distinction concrete.

The screen we propose rests on a single geometric mechanism. Transform each continuous covariate to $U_j = F_j(X_j)$ and let $\xi_j(u)$ denote the tail index of Y given $U_j = u$ alone. Conditioning on one coordinate leaves the others free to range over the fibre $\{u\} \times [0, 1]^{p-1}$, so the projected distribution is a mixture, and a mixture of heavy tails is governed by its heaviest component. Under a full-support condition,

$$\xi_j(u) = \max_{v \in [0, 1]^{p-1}} \gamma(u, v) : \quad (2)$$

the projected index is the upper envelope of the tail-index surface along the fibre. If coordinate j is inactive, every fibre reaches the global maximum $\gamma^* = \max_x \gamma(x)$ and the envelope is flat at γ^* . If coordinate j is active and some of its fibres miss the global argmax set, the envelope dips below γ^* there. Averaging the envelope over a trimmed interval gives a score Ψ_j whose gap $\Delta_j = \gamma^* - \Psi_j$ is zero for every inactive coordinate and positive for every detectable active one. Ranking coordinates by increasing Ψ_j is the screen.

The same geometry states the method's limitation. An active coordinate is invisible when every one of its fibres still meets the global argmax set; this happens, for instance, for $\gamma(u_1, u_2) = \gamma_0 - (u_1 - u_2)^2$, whose argmax is the diagonal. Detectability is a property of the argmax projections, and we characterize it exactly rather than assuming a generic signal-strength condition.

Estimation follows the envelope. For each coordinate we slide a window over the empirical marginal ranks, apply a Hill statistic to the upper α -fraction of the responses in the window, and average the resulting profile. Empirical ranks give every coordinate the same deterministic window geometry and remove the marginal scales. The price is a rank-displacement term, which we control simultaneously over p coordinates by a Dvoretzky–Kiefer–Wolfowitz bound; it is negligible when $\log(pn) = o(nh^2)$. The main result bounds the maximal score error by

$$O_{\mathbb{P}} \left(\sqrt{\frac{\log(pn)}{n\alpha h}} \right) + o(\Delta_{\min}),$$

where $\Delta_{\min}$ is the weakest population gap, and yields sure screening when $\log(pn) = o(n\alpha h\Delta_{\min}^2)$: ultra-high dimension is affordable exactly when the local extreme count dominates the logarithm of the dimension. A blockwise implementation costs $O(pn \log n)$ time.

Our construction adapts the projected-tail framework of Gardes and Podgorny [9], who estimate a low-dimensional central tail-index subspace by minimizing an average of projected local Hill estimates over projection matrices, in fixed dimension. We keep their projection semantics and local Hill estimator, and change what is needed for screening: the projections are the p coordinates themselves, p may diverge, windows are built from empirical ranks with deterministic local counts, and uniformity over a compact matrix class is replaced by union bounds over coordinates and window states. Their density assumptions on the covariate are not needed here; the empirical-rank construction replaces them. Among screening proposals, Yoshida and Umezu [19] screen on covariate-dependent extreme-value indices through kernel conditional Pickands contrasts in a large-bandwidth regime, and parametric tail-index regression [18], extended to high dimensions with regularization by Sasaki et al. [17], exploits a parametric link when one is available. Tail dimension reduction methods [1, 2, 4, 8] target related but distinct conditional-tail objects.

The theoretical scope deserves a clear statement. The primitive model (1) identifies the projected index (2), but does not by itself provide the uniform projected-quantile regularity needed for growing-dimensional concentration: projection can create additional logarithmic slow variation even from exact Pareto conditionals. We therefore state the required conditions directly on the projected quantiles, establish them on an explicit model class, and leave their derivation from primitive assumptions open.

Section 2 develops the population theory. Section 3 defines the estimator and states the screening results. Section 4 reports a simulation study built around four models whose slowly varying factors are driven by inactive covariates, a tuning study, one-factor sensitivity analyses, and a three-method comparison. Section 5 applies the screen to violent-crime rates in U.S. communities. Section 6 discusses limitations. Proofs are in Appendix A.

2 Population theory

2.1 Model and assumptions

All quantities below may depend on n : the dimension $p = p_n$, the tail-index function γ , the active set A , its size $s = |A|$, and later the bandwidth h , the tail fraction α and the retained size d . The index n is suppressed whenever no ambiguity arises.

Notation. A positive measurable function f is slowly varying at infinity if $f(ty)/f(y) \rightarrow 1$ as $y \rightarrow \infty$ for every $t > 0$, and a survival function $\bar{F}$ is regularly varying with index $-1/\xi < 0$ if $\bar{F}(ty)/\bar{F}(y) \rightarrow t^{-1/\xi}$ for every $t > 0$. Generalized inverses are $F^{-1}(t) = \inf\{y : F(y) \geq t\}$. For $u \in [0, 1]$ and $v \in [0, 1]^{p-1}$, $\iota_j(u, v) \in [0, 1]^p$ denotes the vector with u in coordinate j and v in the remaining coordinates, in order; we abbreviate $\gamma(u, v) = \gamma\{\iota_j(u, v)\}$ when j is clear from context. Norms on $[0, 1]^A$ are Euclidean. All asymptotic statements concern a triangular array indexed by n , and $Z_n = O_{\mathbb{P}}(a_n)$ means $\lim_{C \rightarrow \infty} \limsup_n \mathbb{P}(|Z_n| > Ca_n) = 0$.

The observed data are i.i.d. pairs (X_i, Y_i) with $Y_i > 0$ and $X_i \in \mathbb{R}^p$ with continuous marginal distribution functions $F_1, \dots, F_p$. The theory is formulated on the latent probability-integral transforms $U_{ij} = F_j(X_{ij})$, so that $U_i \in [0, 1]^p$ has uniform one-dimensional marginals and unrestricted cross-coordinate dependence; the transforms are never needed by the algorithm, because the ranks of $X_{1j}, \dots, X_{nj}$ coincide with the ranks of $U_{1j}, \dots, U_{nj}$ almost surely. With slight abuse of notation we write γ for the tail-index function on the cube, $\gamma(u) = \gamma_X\{F_1^{-1}(u_1), \dots, F_p^{-1}(u_p)\}$, where γ_X is the function appearing in (1).

(C1) *Conditional regular variation.* For every u , the conditional survival function factorizes as

$$\bar{F}(y \mid u) = \mathbb{P}(Y > y \mid U = u) = y^{-1/\gamma(u)} L(y, u),$$

where $L(\cdot, u)$ is slowly varying at infinity, $\gamma : [0, 1]^p \rightarrow (0, \infty)$ is continuous, and $0 < \underline{\gamma} \leq \gamma(u) \leq \bar{\gamma} < \infty$ uniformly in n and u .

(C2) *Uniform tail regularity.* There is a function c with $0 < c_- \leq c(u) \leq c_+ < \infty$ such that $\sup_u |L(y, u)/c(u) - 1| \rightarrow 0$ as $y \rightarrow \infty$.

(C1) states that each conditional law is heavy-tailed with index $\gamma(u)$, as in (1); (C2) strengthens the slow variation of L to uniform convergence toward a bounded positive limit, the form needed to control mixtures of the conditional laws.

The active set A collects the coordinates on which γ genuinely depends: $j \in A$ if and only if there exist u, v differing only in coordinate j with $\gamma(u) \neq \gamma(v)$. Iterating over the complementary coordinates shows that $\gamma(u) = g(u_A)$ for some function g , and that A is the smallest coordinate set with this property. Membership in A concerns the limiting exponent only, so a coordinate may shift the body of the conditional distribution, or its second-order behaviour, without being active. Conditions (C1)–(C2) parallel conditions (C.1)–(C.2) of Gardes and Podgorny [9], except that we do not assume the law of $\gamma(U)$ to be absolutely continuous; the projection argument below handles atoms, which arise as soon as one coordinate is inactive.

Fix a trimming parameter $0 < \varepsilon < 1/2$ and write $I_\varepsilon = [\varepsilon, 1 - \varepsilon]$. For each coordinate j , fix a pointwise conditional distribution $K_j(u, dv)$ of U_{-j} given $U_j = u$, and define the conditional law of Y given $U_j = u$ to be compatible with it through the disintegration identity

$$\mathbb{P}(Y > y \mid U_j = u) = \int \mathbb{P}\{Y > y \mid U = (u, v)\} K_j(u, dv);$$

without such pointwise versions the statements below hold for almost every u . Define

$$\xi_j(u) = \text{ess sup}_{v \sim K_j(u, \cdot)} \gamma(u, v),$$

the essential supremum of $v \mapsto \gamma(u, v)$ under $K_j(u, \cdot)$; Proposition 2.1 shows that the conditional law of Y given $U_j = u$ is regularly varying with index $-1/\xi_j(u)$, so $\xi_j(u)$ is its tail index. One further condition is used only where stated:

(S) *Full fibre support.* For every j and $u \in I_\varepsilon$, the support of $K_j(u, \cdot)$ is all of $[0, 1]^{p-1}$. A continuous joint density, positive on the interior cube, is sufficient.

2.2 Projection turns the surface into an envelope

Conditioning on $U_j = u$ mixes the conditional laws of Y over the fibre $\{u\} \times [0, 1]^{p-1}$. Heuristically, the slowest-decaying component of a mixture of heavy tails dictates the tail of the mixture, so the projected index should equal the largest tail index available on the fibre, and a localized region of heavy tails should survive projection intact. The following proposition makes this precise.

For a sub- σ -field $\mathcal{G}$, the conditional essential supremum $\text{ess sup}\{\gamma(U) \mid \mathcal{G}\}$ is the smallest $\mathcal{G}$ -measurable random variable dominating $\gamma(U)$ almost surely.

Proposition 2.1 (Projection). *Assume (C1)–(C2) and let $\mathcal{G} \subseteq \sigma(U)$. Then $\mathbb{P}(Y > y \mid \mathcal{G})$ is almost surely regularly varying with index $-1/\xi$, where $\xi = \text{ess sup}\{\gamma(U) \mid \mathcal{G}\}$. In particular $\xi_j(u)$ is the conditional essential supremum of $\gamma(U)$ given $U_j = u$. Under (S),*

$$\xi_j(u) = \max_{v \in [0, 1]^{p-1}} \gamma(u, v), \quad u \in I_\varepsilon. \quad (3)$$

The essential-supremum identity requires neither full support nor absolute continuity; (S) enters only to replace the essential supremum by the maximum over the whole fibre. If dependence removes part of a fibre, the maximum runs over the conditional support actually available at u , and the population target changes accordingly.

2.3 Score, gap, and detectability

Write $\gamma^* = \max_u \gamma(u)$ for the global maximum and

$$\Psi_j = \frac{1}{1 - 2\varepsilon} \int_{I_\varepsilon} \xi_j(u) du, \quad \Delta_j = \gamma^* - \Psi_j, \quad \Delta_{\min} = \min_{j \in A} \Delta_j. \quad (4)$$

The screen ranks coordinates by increasing Ψ_j and keeps the d smallest. Small scores indicate activity: all inactive coordinates share the flat baseline γ^* , while fixing a detectable active coordinate lowers the attainable envelope. The algorithm uses only the ordering of the Ψ_j and never estimates γ^* .

Let $M = \arg \max_u \gamma(u)$, a nonempty compact set, and let $\pi_j(M)$ be its projection onto coordinate j .

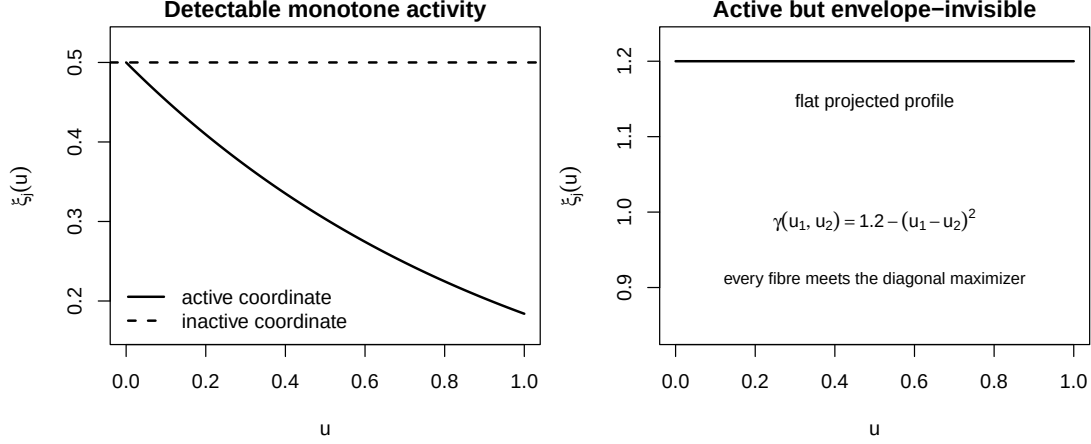


Figure 1: Projected envelopes. Left: a monotone active coordinate depresses the envelope below the inactive baseline γ^* . Right: in the diagonal example every fibre meets the argmax set, so the envelope of an active coordinate stays flat.

Proposition 2.2 (Detectability). *Assume (C1)–(C2) and (S).*

1. If $j \notin A$, then $\xi_j \equiv \gamma^*$ on I_ε and $\Delta_j = 0$.
2. For every j and $u \in I_\varepsilon$: $\xi_j(u) = \gamma^*$ if and only if $u \in \pi_j(M)$.
3. $\Delta_j > 0$ if and only if $I_\varepsilon \setminus \pi_j(M)$ has positive Lebesgue measure.

If the restriction of γ to the active coordinates has a finite set of maximizers, every active coordinate is detectable.

Claim 3 separates two failure modes that finite-sample experiments tend to blur. When $\pi_j(M)$ covers I_ε , no sample size makes coordinate j identifiable from this score; the diagonal example $\gamma(u_1, u_2) = \gamma_0 - (u_1 - u_2)^2$, with $M = \{(t, t) : 0 \leq t \leq 1\}$, is of this kind, and both of its active coordinates are invisible. When $\pi_j(M)$ misses a set of positive measure, detection is a question of estimation accuracy, quantified by Section 3. Figure 1 contrasts the two situations.

Proposition 2.3 (Quantitative margin). *Assume (C1)–(C2), (S), and that for some $u_A^* \in [0, 1]^A$, $c > 0$, $\nu > 0$,*

$$\gamma(u) \leq \gamma^* - c \|u_A - u_A^*\|^\nu \quad \text{for all } u.$$

Then, for $j \in A$,

$$\gamma^* - \xi_j(u) \geq c |u - u_j^*|^\nu, \quad \Delta_j \geq c \mathbb{E}(|V - u_j^*|^\nu), \quad V \sim \text{Unif}(I_\varepsilon).$$

For $\nu = 2$ the right-hand side is at least $c\{(1 - 2\varepsilon)^2/12 + (1/2 - u_j^*)^2\}$.

Two consequences of (3) help with examples. If $\gamma(u) = G(\sum_{\ell \in A} g_\ell(u_\ell))$ with G strictly increasing, then $\xi_j(u) = G(g_j(u) + \sum_{\ell \in A, \ell \neq j} \max g_\ell)$, and $\Delta_j > 0$ whenever g_j lies below its maximum on a subset of I_ε of positive measure. If $\partial_j \gamma \geq c_j > 0$ throughout the cube, arbitrary interactions are allowed and $\Delta_j \geq c_j/2$.

For the benchmark $\gamma(u) = \frac{1}{2} \exp(-u_1 - u_2 - u_3 - u_4)$ used in Section 4, $\xi_j(u) = e^{-u}/2$ for $j \leq 4$ and $\xi_j \equiv 1/2$ otherwise, so $\Delta_j \approx 0.186$ at $\varepsilon = 0.05$. The gap is comfortable, yet estimating $\xi_j(u)$ near its maximum requires sample points whose remaining three active coordinates are simultaneously small; the simulations keep population detectability and finite-sample accessibility clearly separated.

3 Estimation and screening theory

3.1 Empirical-rank local Hill scores

The estimator follows the envelope construction coordinate by coordinate. Let R_{ij} be the rank of X_{ij} among $X_{1j}, \dots, X_{nj}$ and $\widehat{U}_{ij} = R_{ij}/(n+1)$. For $u \in I_\varepsilon$, form the rank window

$$\mathcal{W}_j(u) = \{i : |\widehat{U}_{ij} - u| \leq h\}, \quad M_j(u) = |\mathcal{W}_j(u)|, \quad k_j(u) = \lfloor \alpha M_j(u) \rfloor,$$

and, writing $Y_{j,u,(1)} \leq \dots \leq Y_{j,u,(M_j(u))}$ for the local order statistics, the local Hill estimate

$$\widehat{\xi}_j(u) = \frac{1}{k_j(u)} \sum_{r=1}^{k_j(u)} \log \frac{Y_{j,u,(M_j(u)-r+1)}}{Y_{j,u,(M_j(u)-k_j(u))}}, \quad (5)$$

defined when $\alpha M_j(u) > 1$. This is the local-fraction Hill statistic of Gardes and Podgorny [9, Definition 5] evaluated on one-coordinate rank windows. The score averages the profile over the deterministic grid $G = \{r/(n+1) : r = 1, \dots, n\} \cap I_\varepsilon$,

$$\widehat{\Psi}_j = \frac{1}{|G_j^{\text{val}}|} \sum_{u \in G_j^{\text{val}}} \widehat{\xi}_j(u), \quad G_j^{\text{val}} = \{u \in G : \alpha M_j(u) > 1\}, \quad (6)$$

with $\widehat{\Psi}_j = +\infty$ when no grid point is usable, so that underpopulation can never favour a coordinate. Because the window counts $M_j(u)$ depend on the ranks only through the grid position, G_j^{val} is in fact the same set for every j ; under the conditions of Theorem 3.1 every grid point is usable and $G_j^{\text{val}} = G$. The screened set $\widehat{A}_d$ collects the d smallest scores, ties broken by coordinate index.

The two localization parameters play different roles: h fixes how many observations resemble the target rank position, roughly $2nh$ of them, and α fixes how many of their largest responses enter the Hill statistic, roughly $2n\alpha h$. Rank windows make these counts deterministic up to boundary effects, which is what permits exact finite-sample union bounds over a diverging number of coordinates.

3.2 Assumptions

Ranks displace the unobserved transforms U_{ij} , windows displace the conditioning point, and the Hill statistic sees only the upper α -fraction of a projected distribution. The assumptions bound these three effects uniformly over the coordinates. Throughout this section every active coordinate is assumed detectable, $\Delta_{\min} > 0$ for every n , so that the conditions below are well posed. Let $Q_j(\cdot, u)$ denote the generalized upper conditional quantile of Y given $U_j = u$, $Q_j(s, u) = F_{Y|j}^{-1}(1 - s^{-1} | u)$ with $F^{-1}(t) = \inf\{y : F(y) \geq t\}$, and let $\tau_n = (pn)^{-2}$.

(E1) *Quantile representation and regularity of the projected index.* For every $j \leq p$ and every $u \in [0, 1]$ the conditional distribution function $F_{Y|j}(\cdot | u)$ is continuous, so that $V_{ij} = 1 - F_{Y|j}(Y_i | U_{ij})$ is conditionally standard uniform and $Y_i = Q_j(1/V_{ij}, U_{ij})$ almost surely. Moreover the projected indices are equicontinuous on I_ε , uniformly over coordinates and over n ,

$$\omega_\xi(r) = \max_{j \leq p} \sup_{u, v \in I_\varepsilon, |u-v| \leq r} |\xi_j(u) - \xi_j(v)| \longrightarrow 0 \quad (r \rightarrow 0),$$

and satisfy the grid-scale regularity condition $\omega_\xi(1/n) = o(\Delta_{\min})$.

(E2) *Uniform tail regularity.* Writing $Q_j(s, u) = s^{\xi_j(u)} \ell_j(s, u)$ with $\ell_j(\cdot, u)$ slowly varying,

$$\max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{s \geq (3\alpha)^{-1}, t > 1} \frac{|\log\{\ell_j(ts, u)/\ell_j(s, u)\}|}{\log t} = o(\Delta_{\min}).$$

(E3) *Uniform local quantile regularity.* There exists a constant $C_0 > 1$ such that

$$\max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{v \in [0, 1], |u-v| \leq C_0 h} \sup_{w \in [\tau_n, 1-\tau_n]} \left| \log \frac{Q_j(w^{-1}, v)}{Q_j(w^{-1}, u)} \right| = o(\Delta_{\min}).$$

The theorems use only this one C_0 ; under (7) any $C_0 > 1$ suffices.

(E1) asks for continuity only; strict monotonicity of the conditional distribution function is not needed, because flat pieces are visited with probability zero. (E2) is the analogue of condition (H.4) in Gardes and Podgorny [9], normalized by $\log t$ and required only beyond the working tail level $(3\alpha)^{-1}$; it accommodates slowly varying factors such as $\ell(s) \asymp (\log s)^{-2}$, for which the left side is of order $1/\log(1/\alpha) \asymp 1/\log n$ when $\alpha = n^{-a}$, so gaps $\Delta_{\min} \gg 1/\log n$ are covered and smaller gaps are not. (E3) adapts their condition (H.5): the truncation $\tau_n = (pn)^{-2}$ makes the probability that any of the pn conditional PIT variables leaves $[\tau_n, 1 - \tau_n]$ vanish, which is the high-dimensional replacement for their fixed-dimension truncation. Their density conditions (H.1)–(H.2) have no counterpart here: empirical-rank windows have deterministic counts, so no lower bound on a covariate density is required.

These conditions are stated on the projected quantiles, not derived from (C1)–(C2): projection can generate genuinely new slow variation. If $W \sim U(0, 1)$ and $\mathbb{P}(Y > y \mid W) = y^{-1/(1+W)}$ exactly, then $\mathbb{P}(Y > y) \sim 4y^{-1/2}/\log y$, so even exact Pareto conditionals produce a logarithmic factor after projection. Appendix A.6 establishes (E1)–(E3) for finite mixtures with a spectral gap, where the left side of (E2) is $O(\alpha^\rho)$, and sketches the same for a dependent plateau-maximizer class where it is $O(\{\log(1/\alpha)\}^{-\{1+(s-1)/a_0\}})$, a_0 being the flatness of the plateau boundary; the least favourable case $s = 2$ gives exponent $1 + 1/a_0$.

3.3 Main results

Theorem 3.1 (Score concentration). *Assume (C1)–(C2), (S), (E1)–(E3), $\alpha \leq 1/3$, $h \leq 1 - \varepsilon$, and*

$$\log(pn) = o(nh^2), \quad \log(pn) = o(n\alpha h). \quad (7)$$

Then there are deterministic sequences $a_n = O(\sqrt{\log(pn)/(n\alpha h)})$ and $b_n = o(\Delta_{\min})$ with

$$\mathbb{P}\left(\max_{j \leq p} |\hat{\Psi}_j - \Psi_j| \leq a_n + b_n\right) \longrightarrow 1;$$

in shorthand, $\max_{j \leq p} |\hat{\Psi}_j - \Psi_j| = O_{\mathbb{P}}(\sqrt{\log(pn)/(n\alpha h)}) + o(\Delta_{\min})$.

The stochastic term is the exponential-spacings fluctuation of $O(n)$ local Hill statistics per coordinate, each built from at least $c n \alpha h$ spacings, controlled simultaneously over p coordinates; $\sqrt{\log(pn)/(n\alpha h)}$ is the resulting effective-tail-sample-size rate. The $o(\Delta_{\min})$ term collects the tail bias (E2), the spatial bias (E3) evaluated at $C_0 h$ — the first condition in (7) makes the rank displacement smaller than $(C_0 - 1)h$, so windows never leave the range covered by (E3) — and the quadrature error $\omega_\xi(1/n) + O(1/n)$ from (E1).

Theorem 3.2 (Sure screening). *Assume the conditions of Theorem 3.1 (including $\Delta_{\min} > 0$, i.e. every active coordinate detectable in the sense of Proposition 2.2) and*

$$\log(pn) = o(n\alpha h \Delta_{\min}^2). \quad (8)$$

Then, for every $d \geq s$,

$$\mathbb{P}(A \subseteq \hat{A}_d) \rightarrow 1,$$

and with probability tending to one every active score is smaller than every inactive score.

Corollary 3.3 (Exact recovery). *Under the conditions of Theorem 3.2, if $d = s$ then $\mathbb{P}(\hat{A}_s = A) \rightarrow 1$.*

The proof passes through a uniform bound on the whole profile $\max_j \sup_{u \in I_\varepsilon} |\hat{\xi}_j(u) - \xi_j(u)|$, stated with explicit finite-sample constants as Theorem A.12 in the appendix; Theorems 3.1 and 3.2 are its asymptotic readings. The argument has four ingredients. A single Dvoretzky–Kiefer–Wolfowitz event per coordinate controls all rank displacements at once, contributing the $\log(pn) = o(nh^2)$ requirement. Rank windows have deterministic sizes, at least $\lfloor \{h + \min(h, \varepsilon)\}(n+1) \rfloor - 1$ observations each, and, as u moves, pass through at most $4n+1$ membership states per coordinate. Conditionally on a covariate column, the probability-integral transforms of the responses are i.i.d. uniform, and the Rényi representation turns each local Hill statistic into a mean of i.i.d. exponentials plus a slowly varying remainder bounded through (E2). A union bound over the $p(4n+1)$ states then gives the $\log(pn)$ factor; no independence across overlapping windows or dependent coordinates is used. Finally, score separation is deterministic: when every score error is below $\Delta_{\min}/2$, every active score lies below $\gamma^* - \Delta_{\min}/2$ and every inactive score above it. Gardes and Podgornyy [9, Theorem 3] establishes the corresponding uniform consistency in fixed dimension over a compact class of projection matrices; the present setting replaces their covering-number argument and density assumptions by the rank construction above, and their result does not imply the growing- p bound needed here.

Condition (8) quantifies the cost of dimension: screening p coordinates costs $\log p$ against the effective local extreme count $n\alpha h$, deflated by the squared gap. With $\alpha = n^{-a}$ and $h = n^{-b}/2$ and $b \leq a$, dimensions up to $\log p = o(n^{1-a-b} \Delta_{\min}^2)$ are admissible, a polynomial range of p for fixed gaps; for $b > a$ the rank-displacement condition $\log(pn) = o(nh^2) = o(n^{1-2b})$ binds first. These ranges treat (E3) as given: in the verified mixture class of Proposition A.16 the spatial modulus adds the constraint $\log(pn) = o(n^b)$, giving the range $\log(pn) = o(n^{\min(b, 1-a-b)})$ stated there.

4 Simulation study

Each coordinate is scored in $O(n \log n)$ time: one sort by empirical rank, then a sliding window over the rank grid whose local Hill statistics are maintained by a Fenwick tree indexed by global response ranks. Covariates are generated and processed in blocks, so the total cost is $O(pn \log n)$ time and $O(n)$ working memory beyond the scores. At $(n, p) = (2000, 1000)$ one complete screen takes under a second on a laptop core. Scores are sorted increasingly, ties broken by coordinate index; a coordinate whose grid is entirely underpopulated receives score $+\infty$.

4.1 Models

All four models share the covariate design and the active set $A = \{1, 2, 3, 4\}$. Latent Gaussian AR(1) covariates $Z_{ij} = \rho Z_{i,j-1} + \sqrt{1 - \rho^2} \epsilon_{ij}$ are transformed to $U_{ij} = \Phi(Z_{ij})$; the tail-index functions are

evaluated at the population transforms U , while the estimator sees only Z and computes its own empirical ranks. Responses are generated by quantile inversion,

$$Y = V^{-\gamma(U)} \ell(1/V, U), \quad V \sim \text{Unif}(0, 1),$$

the mechanism used by Gardes and Podgorny [9]. We combine three tail-index functions from increasing difficulty,

$$\begin{aligned} \gamma_1(u) &= 0.5 \exp\left(-\sum_{j=1}^4 u_j\right), \\ \gamma_2(u) &= 0.5 \exp\left(-0.5 \sum_{j=1}^4 u_j\right), \\ \gamma_3(u) &= 0.55 \exp\left(-0.35 \sum_{j=1}^4 u_j - 0.20 u_1 u_2 - 0.15 u_3 u_4\right), \end{aligned}$$

with two slowly varying functions in the forms of Gardes and Podgorny [9, Section 4.3],

$$\ell^{(1)}(s, u) = [1 + \exp\{v(u) - s\}]^{-1}, \quad \ell^{(2)}(s, u) = e^{-1/(2s)} v_\kappa(u),$$

where $v(u) = 2.5(u_5 + u_6 + u_7 + u_8) \in (0, 10)$ matches the range of their $B_1^\top x$, and the scale factor of $\ell^{(2)}$ is

$$v_\kappa(u) = \exp\left\{\kappa \sum_{j=5}^8 \left(u_j - \frac{1}{2}\right)\right\} \in [e^{-2\kappa}, e^{2\kappa}],$$

strictly positive, bounded, symmetric in the four *inactive* variables, and controlled by the single strength parameter κ ; its bounds keep the model compatible with the uniform constants in (C2). The four models are

$$\text{M1} = (\gamma_1, \ell^{(1)}), \quad \text{M2} = (\gamma_1, \ell_{\kappa^*}^{(2)}), \quad \text{M3} = (\gamma_2, \ell^{(1)}), \quad \text{M4} = (\gamma_3, \ell^{(1)}).$$

M1 is the reference. M2 keeps the tail index of M1 and changes only the slowly varying factor: $\ell^{(2)}(s, u) \rightarrow v_\kappa(u)$ as $s \rightarrow \infty$, so the tail index depends on $A_\gamma = \{1, 2, 3, 4\}$ while the scale — hence every high finite quantile — also depends on $A_{\text{scale}} = \{5, 6, 7, 8\}$. Variables can therefore matter for a 0.95-quantile screen without entering the limiting exponent, by construction; Section 4.4 exploits this. The strength $\kappa^* = 0.20$ was chosen by a pilot on a dedicated seed stream, never reused below: candidate values $\kappa \in \{.15, .25, .35, .50, .70\}$, plus the single authorized midpoint extension $\kappa = .20$, were evaluated on 80 replications each. The selection rule, fixed in the pilot code before inspection, required the smallest κ with (i) tail-screen Sure-20 at least 0.5, (ii) clearly more scale variables in the quantile screen's top four than in the tail screen's, (iii) more of A_γ in the tail screen's top four than in the quantile screen's, and (iv) at least one scale variable in the quantile screen's top four on average. No candidate satisfied all four: at $\kappa = .20$, criteria (i), (ii) and (iv) hold (Sure-20 0.55, with standard error 0.056 on 80 replications, so (i) itself clears its threshold only within one standard error; 1.14 against 0.17 scale variables) but (iii) fails by a margin within Monte Carlo error (2.71 against 2.83 of A_γ), while at $\kappa = .25$ criterion (i) fails (0.40). We therefore applied the protocol's declared fallback — the candidate best balancing scale separation against tail estimability — and fixed $\kappa^* = 0.20$; the choice is a judgment call informed by the pilot, not the output of a rule that selected uniquely. Full pilot results are in the supplementary table (`pilot_summary.csv`). M3 halves the exponent range of the signal, weakening the population gaps from 0.186 to 0.107; M4 adds interactions among the active coordinates while remaining detectable (Proposition 2.2), with trimmed gap 0.086.

None of the four models belongs to the sufficient classes of Appendix A.6: their tail-index surfaces are continuous with maximizers at a corner of the active cube, so the fibre-maximizer

laws have no atom and the projected quantiles acquire slow variation of logarithmic type — the boundary regime in which condition (E2) covers only gaps large relative to $1/\log n$. At $n = 2000$ the corresponding bias modulus is of order one, larger than every population gap (0.086–0.186), so the finite-sample bounds of Section 3 are not numerically informative for these designs. The simulations therefore probe the method beyond the regime its theory certifies. Ranking nevertheless succeeds because much of the local Hill bias is shared across coordinates — every inactive coordinate estimates the same projected mixture — and the screen uses only score differences; the experiments should be read as evidence about this common-mode behaviour, which the theory does not exploit, rather than as confirmation of the bounds.

Unless a factor is explicitly varied, the design is

$$n = 2000, \quad p = 1000, \quad \rho = 0.25, \quad d = 20, \quad \varepsilon = 0.05,$$

with 200 Monte Carlo replications per cell; $\text{Sure-}d$ denotes $\mathbb{P}(R_{\max} \leq d)$ with $R_{\max} = \max_{j \in A} \text{rank}(j)$, and binomial standard errors are at most 0.036.

4.2 Tuning

The two smoothing parameters are parameterized as

$$\alpha = n^{-a}, \quad h = n^{-b}/2,$$

following the convention of Gardes and Podgorny [9]. We evaluate $\text{Sure-}20$ for each of the four models over the crossed grid $a \in \{.25, .30, .35, .40, .45, .50\}$, $b \in \{0, .05, .10, .15, .20, .30, .40\}$ at the baseline design; Figure 2 shows the four heatmaps.

Three features stand out. The row $a = 0.25$ collapses for the three $\ell^{(1)}$ models ($\text{Sure-}20$ at most .115, .085 and .060 for M1, M3 and M4) but not for M2 (up to .475): at $\alpha = n^{-0.25} \approx 0.15$ the local Hill statistics reach into quantile regions where the factor $\ell^{(1)}$, whose argument $v(u) - s$ is dominated by the U_5 – U_8 combination $v(u) \in (0, 10)$ at moderate s , deforms the projected distributions; the recalibrated $\ell^{(2)}$, bounded within $[e^{-0.4}, e^{0.4}]$, produces no such deformation. Performance also degrades toward large a and large b together, where the local extreme count collapses, and the degenerate bandwidth $b = 0$, which retains half of the rank scale in every window, is never optimal. In between, the row $a = 0.30$ performs best for M1, M3 and M4 simultaneously over $b \in \{.05, \dots, .20\}$, with $\text{Sure-}20$ ranges (.915–.975, .445–.605, .300–.385). The M1 variation over this region is small (0.06, within two to three standard errors of a difference); the M3 and M4 variations are larger (up to 0.16, about three standard errors of a difference), so the region is one of good rather than statistically indistinguishable performance, and we select the value of b at which both M3 and M4 attain their grid maxima. We fix

$$(a^*, b^*) = (0.30, 0.10), \quad \text{i.e.} \quad \alpha = n^{-0.3}, \quad h = n^{-0.1}/2,$$

for every subsequent experiment, in simulation and in the application; at $n = 2000$ this gives $\alpha \approx 0.10$, $h \approx 0.23$, and about $2nah \approx 96$ effective local extremes per interior window, whose width is $2h$. No model is given its own tuning.

M2 behaves like a damped M1: its $\text{Sure-}20$ reaches 0.555 at (a^*, b^*) and its grid maximum, 0.625, sits at the deeper tail fraction $a = 0.40$ with the same $b = 0.10$ — the direction predicted by condition (E2), since a smaller α reduces the bias contributed by $\ell^{(2)}$. M2 uses the common tuning, at a cost of 0.07 relative to its own best cell; its role in Section 4.4 is to separate statistical targets.

Two caveats on this selection protocol apply throughout. $\text{Sure-}20$ is an oracle criterion — it requires knowing the active set — so the choice of (a^*, b^*) is a simulation-calibrated default, not a

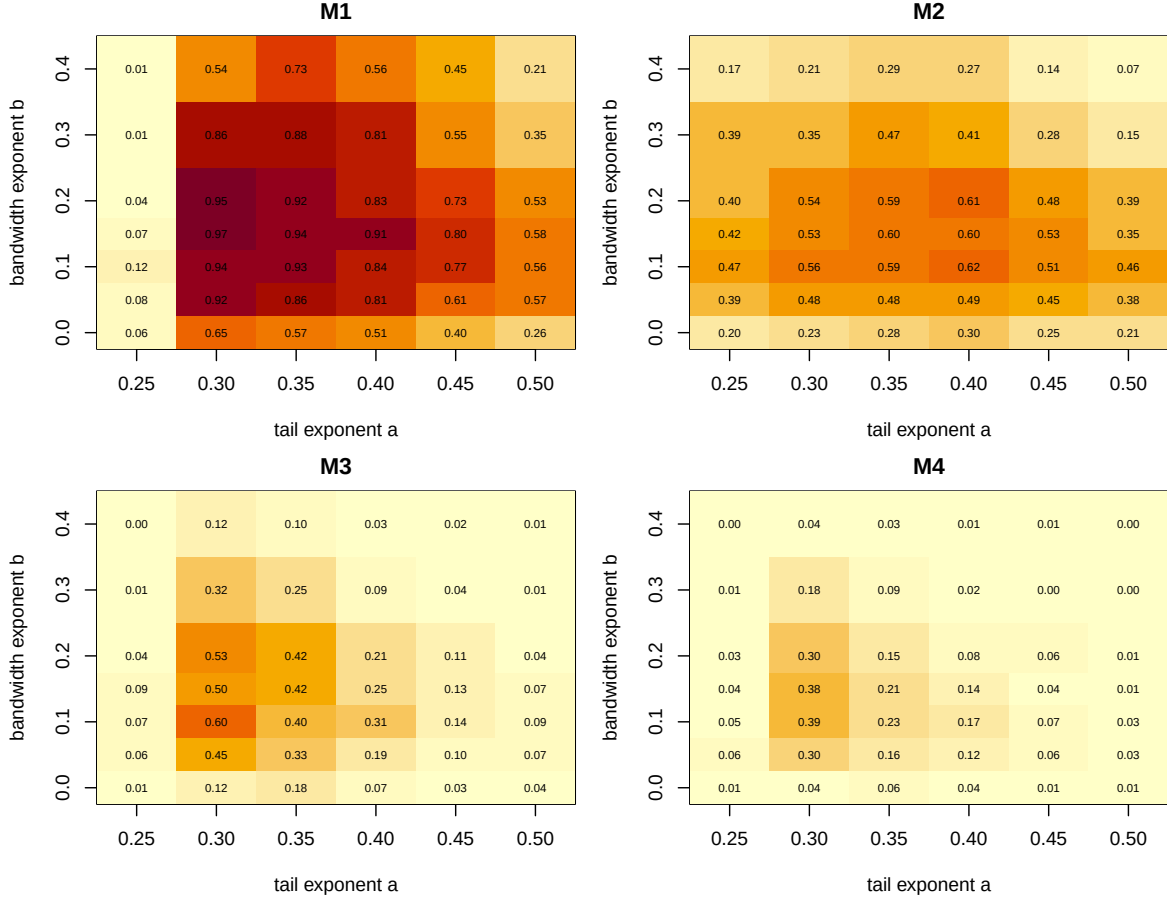


Figure 2: Sure-20 over the (a, b) tuning grid for the four models at $n = 2000$, $p = 1000$, $\rho = 0.25$ (200 replications per cell; $\alpha = n^{-a}$, $h = n^{-b}/2$). Standard errors are at most 0.036.

data-driven rule; no such rule is proposed in this paper, and the half-sample stability diagnostics of Section 5 are the practical substitute. Second, for the $\ell^{(1)}$ models $a^* = 0.30$ sits directly above the collapsed row $a = 0.25$, so moderate misspecification of a toward larger tail fractions is the dangerous direction there — model-dependently, since M2 is robust at $a = 0.25$; the real-data stability grid includes $a = 0.25$ deliberately.

4.3 Influence of the design factors

This subsection uses Model M1 only, varies one factor at a time around the baseline $(n, p, \rho, d) = (2000, 1000, 0.25, 20)$, on a separate seed stream from the tuning study. At the baseline, Sure-20 is 0.950 and the mean worst active rank is 9.2; the same design cell appears in the tuning grid (0.935) and in the comparison below (0.925), each from its own replications, and the three values agree within Monte Carlo error. Figure 3 displays the three one-factor curves.

Sample size acts through the local extreme count, in the direction the rate suggests: Sure-20 rises from 0.400 at $n = 1000$ to 0.950 at $n = 2000$ and 1.000 at $n = 5000$. The ambient dimension degrades performance slowly: 0.995, 0.960, 0.950, 0.905 at $p = 100, 500, 1000, 5000$ — a five-fold increase of p beyond the baseline costs less than five points of screening probability, consistent with the logarithmic entry of p in Theorem 3.2. Dependence helps in this design: Sure-20 is 0.615 at

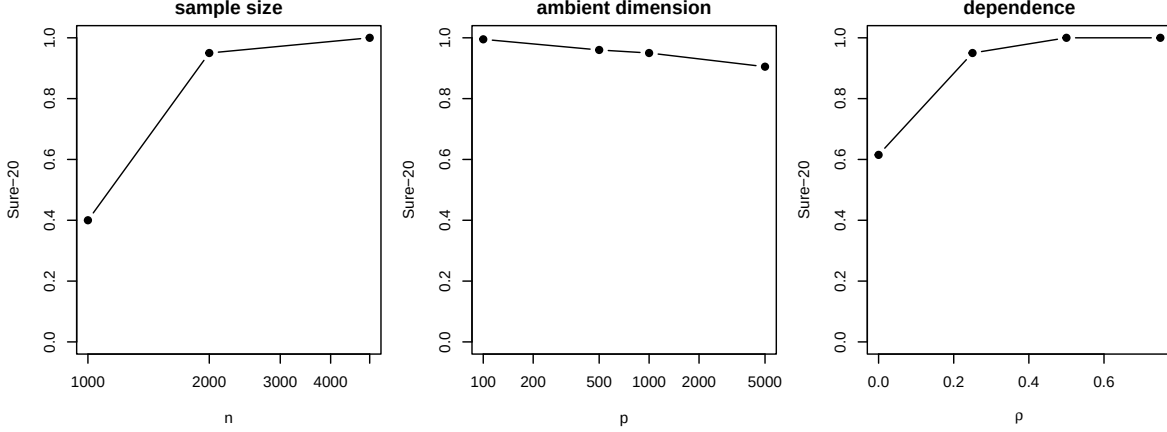


Figure 3: One-factor sensitivity for Model M1 around the baseline $(n, p, \rho) = (2000, 1000, 0.25)$, $d = 20$, at the common tuning $(a^*, b^*) = (0.30, 0.10)$; 200 replications per point (standard errors at most 0.036).

$\rho = 0$ and reaches 1.000 at $\rho \geq 0.5$. The active block $\{1, 2, 3, 4\}$ is contiguous in the AR(1) chain, so positive correlation concentrates observations near the diagonal of the active cube, where the envelope is estimated; the direction of this effect is design-specific and correlated inactive neighbours of the active block can equally be promoted in finite samples.

Figure 4 shows $d \mapsto \mathbb{P}(R_{\max} \leq d)$ at the baseline; the ranking underlying the curve does not depend on d . The curve starts at 0.805 for $d = 4$, which is exact recovery since $|A| = 4$, passes 0.950 at the default $d = 20$, and reaches 0.990 by $d = 100$. A moderate retained dimension therefore buys near-certain inclusion, which is the intended reading of a screening guarantee.

4.4 Comparison with competing screens

Three methods are compared on the same 200 simulated datasets per model at the baseline design: the proposed tail-index screen with the tuning (a^*, b^*) fixed in Section 4.2; the conditional-Pickands screen of Yoshida and Umezu [19], implemented from their equations (2)–(4) with the Epanechnikov kernel and the tuning of their numerical work transplanted proportionally ($h = 1$, $k = \lfloor 0.072n \rfloor$), on 25 equally spaced evaluation points in $[.02, .98]$; the bandwidth sensitivity below varies h only, with k kept at their ratio; and the quantile-adaptive SIS of He et al. [11], using cubic B-spline marginal quantile regression with three degrees of freedom, identical at every quantile level, its utility being the empirical squared deviation of the fitted conditional quantile from the marginal one. The two competitors rank large utilities first; ours ranks small scores first. Table 1 reports the levels $\tau = .95$ and $\tau = .99$; Section 4.5 varies τ over $\{.90, .95, .975, .99\}$ on the same datasets.

At $\tau = .95$, quantile SIS performs best in M1, M3 and M4 (Sure-20 1.000, .815 and .690 against our .925, .535 and .345), where the monotone γ moves the entire upper distribution and the quantile utility pools all $n = 2000$ observations at a moderately extreme level, while the tail-index statistic works from about $2n\alpha h \approx 96$ local extremes per window. At $\tau = .99$ the ordering reverses on all four models (our .925, .570, .535, .345 against .610, .395, .085, .010). The Pickands-based screen of Yoshida–Umezu trails on all four models at the transplanted tuning; rerunning it at bandwidths $h \in \{0.5, 1.5\}$ on the same datasets does not change its position (best Sure-20 over the three bandwidths: .445, .270, .165, .160, the M2 value at $h = 1.5$).

M2 separates the two statistical targets. Its tail index depends on $A_\gamma = \{1, 2, 3, 4\}$ only, while the

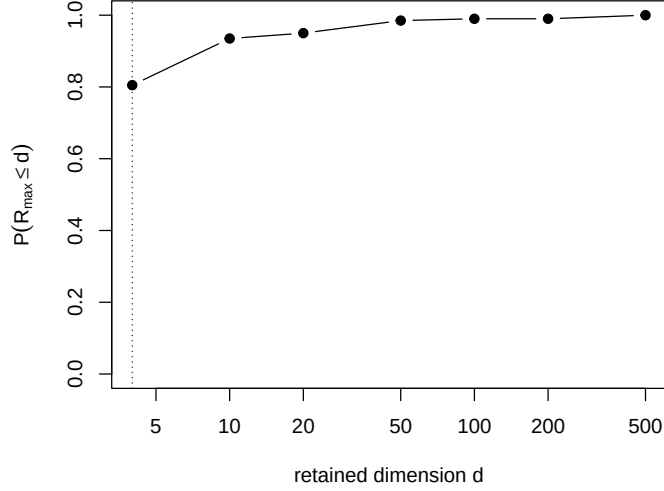


Figure 4: Sure- d curve for Model M1 at the baseline design $(n, p, \rho) = (2000, 1000, 0.25)$, tuning $(a^*, b^*) = (0.30, 0.10)$, 200 replications (standard errors at most 0.036). The value at $d = 4$ is exact recovery, and the ranking underlying the curve does not depend on d .

scale factor v_{κ^*} carries $A_{\text{scale}} = \{5, 6, 7, 8\}$ into every finite quantile. Table 2 shows the composition of each method’s four leading positions. The quantile screen at $\tau = .90$ places on average 2.28 scale variables in its top four — correct behaviour for its own target, since those variables genuinely move the .90 quantile — and at $\tau = .95$ still 1.12. The tail-index screen places 0.16. Both recover A_γ in their top four at nearly the same rate (2.87 for the tail screen, 2.82 for $\tau = .95$), so on M2 the practical difference between the two screens is not power but what the leading positions contain: an analyst reading a $\tau = .95$ ranking as a statement about the limiting tail would carry forward one or two variables that do not determine it. As τ increases the quantile target approaches the tail index and the scale contamination falls (.35 at $\tau = .975$, .03 at $\tau = .99$), but the variance cost rises faster: Sure-20 drops from .940 at $\tau = .95$ to .395 at $\tau = .99$, below our .570.

4.5 Sensitivity of quantile SIS to the level

Figure 5 plots Sure-20 of the quantile screen against $\tau \in \{.90, .95, .975, .99\}$ for the four models, on the same datasets as Table 1. Two forms of sensitivity combine. First, changing τ changes the statistical target $Q_\tau(Y | X)$: under M2, lower levels weight the scale variables more heavily, which is visible in Table 2. Second, raising τ thins the information available for estimating the target: every curve is highest at $\tau = .90$ or .95 and collapses by $\tau = .99$ (M3: .815 at .95, .085 at .99; M4: .690 to .010). The tail-index screen is subject to a bias–variance compromise of the same kind through α , but with a fixed asymptotic target: moving α changes the threshold used to estimate γ , not the definition of γ . Neither method is tuning-free; the tunings mean different things.

The comparison supports a conditional recommendation. For a scientific question about a fixed high quantile, or when high quantiles and the limiting exponent can be expected to share their covariates, the quantile screen at a moderate level exploits the full sample and is the better tool. For a question about the limiting exponent itself — which covariates change the polynomial rate of decay — the tail-index screen targets that quantity directly, keeps scale variables out of its leading positions, and avoids the variance collapse that affects the quantile screen at genuinely extreme levels. Reading the two rankings side by side, as in Section 5, identifies the variables on which the

Table 1: Method comparison at $n = 2000$, $p = 1000$, $\rho = 0.25$, on the same 200 replications per model; the proposed screen uses $(a^*, b^*) = (0.30, 0.10)$, the Yoshida–Umezū screen its published tuning $h = 1$, $k = \lfloor 0.072n \rfloor$ (see the text for a bandwidth grid), and the quantile screens cubic B -splines with three degrees of freedom at every τ . Sure- d is the probability that all four tail-index-active variables rank among the d smallest scores ($d = 4$ is exact recovery); $\mathbb{E}(R_{\max})$ is the mean worst active rank, reported to two decimals. Standard errors of the probabilities are at most 0.036.

Model	Method	Sure-4	Sure-20	$\mathbb{E}(R_{\max})$
M1	Tail-index SIS	0.780	0.925	8.45
	Yoshida–Umezū	0.190	0.410	190.03
	Quantile SIS ($\tau = .95$)	0.870	1.000	4.24
	Quantile SIS ($\tau = .99$)	0.070	0.610	22.89
M2	Tail-index SIS	0.240	0.570	85.02
	Yoshida–Umezū	0.005	0.115	269.21
	Quantile SIS ($\tau = .95$)	0.160	0.940	8.46
	Quantile SIS ($\tau = .99$)	0.005	0.395	38.49
M3	Tail-index SIS	0.230	0.535	103.11
	Yoshida–Umezū	0.010	0.115	346.90
	Quantile SIS ($\tau = .95$)	0.305	0.815	14.98
	Quantile SIS ($\tau = .99$)	0.000	0.085	115.00
M4	Tail-index SIS	0.070	0.345	167.38
	Yoshida–Umezū	0.015	0.100	336.59
	Quantile SIS ($\tau = .95$)	0.135	0.690	28.89
	Quantile SIS ($\tau = .99$)	0.000	0.010	221.90

two questions part ways.

Table 2: Composition of the four leading positions under Model M2 ($\kappa^* = 0.20$), over the same 200 replications: expected number of tail-index-active variables ($A_\gamma = \{1, 2, 3, 4\}$) and of scale variables ($A_{\text{scale}} = \{5, 6, 7, 8\}$) among the four leading positions of each method, rounded half up to two decimals.

Method	$\mathbb{E} \hat{A}_4 \cap A_\gamma $	$\mathbb{E} \hat{A}_4 \cap A_{\text{scale}} $
Tail-index SIS	2.87	0.16
Yoshida–Umezu	0.62	0.75
Quantile SIS ($\tau = .90$)	1.73	2.28
Quantile SIS ($\tau = .95$)	2.82	1.12
Quantile SIS ($\tau = .975$)	3.11	0.35
Quantile SIS ($\tau = .99$)	1.50	0.03

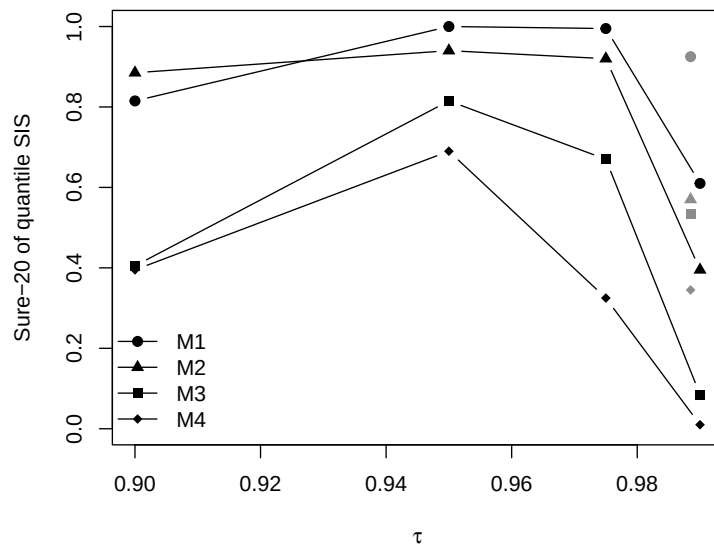


Figure 5: Sure-20 of the quantile-adaptive SIS as a function of the level τ , for the four models at $n = 2000$, $p = 1000$, $\rho = 0.25$ (200 common replications per model; grey symbols at the right edge mark the tail-index screen at the same design). Standard errors are at most 0.036.

5 Application: violent crime across U.S. communities

The analysis contains $n = 1,993$ communities and $p = 99$ candidate predictors. Under the tuning selected in Section 4.2, $\alpha = n^{-0.3} \approx 0.102$ and $h = n^{-0.1}/2 \approx 0.234$, so the sample contains $n\alpha \approx 204$ upper-tail observations in total and about $2nah \approx 95$ within a typical interior rank window (no fewer than about 58 near the trimmed boundary). The dimension is therefore comparable to the number of observations available for local tail estimation. Unrestricted or nonparametric joint tail modelling with 99 covariates is poorly supported by roughly 200 upper-tail observations, and by fewer still within any local covariate neighbourhood; a marginal screen can rank the 99 candidates from exactly this information.

5.1 Data and protocol

The Communities and Crime data set [15], distributed by the UCI Machine Learning Repository in its unnormalized version, links socio-economic census attributes, law-enforcement survey attributes and FBI crime counts for 2,215 U.S. communities. The response is the violent-crime rate per 100,000 inhabitants. We keep the communities with an observed positive response ($n = 1,993$). Preprocessing of the 124 documented predictive attributes is response-blind: we retain numeric predictors with fewer than 5% missing values, which removes the 22 law-enforcement variables reported by only 343 communities, and with at least 100 distinct observed values, which removes 3 low-cardinality variables, keeping the application close to the continuous-marginal framework of the theory. This leaves $p = 99$ continuous predictors — demographic composition, income and poverty, education, employment, family structure, immigration, language, housing, and population density. The at most one missing value per retained predictor is median-imputed, and all three screens below work from the same imputed matrix.

The response is strongly right-skewed: median 374, 99th percentile 2,943, maximum 4,877. The left panel of Figure 6 shows the Hill curve of the response over $k \in [20, 600]$ upper order statistics; it is positive and moves slowly, between 0.283 and 0.374 for $k \in [100, 300]$, with value 0.328 at the working level $k = n\alpha = 204$. The middle panel shows the Pareto QQ plot of the 600 largest responses, approximately linear over the working range with least-squares slope 0.27. The gap between the QQ slope and the Hill value at the same level (0.33) is mild evidence that regular variation is only approximate over the working range. The response exhibits a sufficiently heavy and stable upper tail to make a tail-index analysis plausible; neither diagnostic establishes regular variation beyond doubt, and no single threshold carries a structural interpretation.

The screen uses $\alpha = n^{-0.3}$, $h = n^{-0.1}/2 \approx 0.234$, $\varepsilon = 0.05$. This tuning is the simulation-calibrated default of Section 4.2; it is transferred to these data without any claim that the crime-rate regime resembles the simulation designs, which is why stability rather than a single fit carries the conclusions. Stability is assessed three ways, none of which was used to choose the tuning: the 3×3 grid $a \in \{.25, .30, .35\} \times b \in \{.05, .10, .15\}$, which deliberately includes the tail fraction $a = .25$ found fragile in simulation; 200 seeded random half-samples refit with the primary rule; and 20 randomized breakings of the few exact ties. The two competing screens of Section 4.4 are run on the same imputed matrix.

5.2 Results

Family structure dominates the ranking. The proportion of children in two-parent households (`pctKids2Par`) ranks second in the primary fit, within the top five in every tuning-grid fit, and in the top five of 92% of the 200 half-samples — the most stable variable in the analysis. The

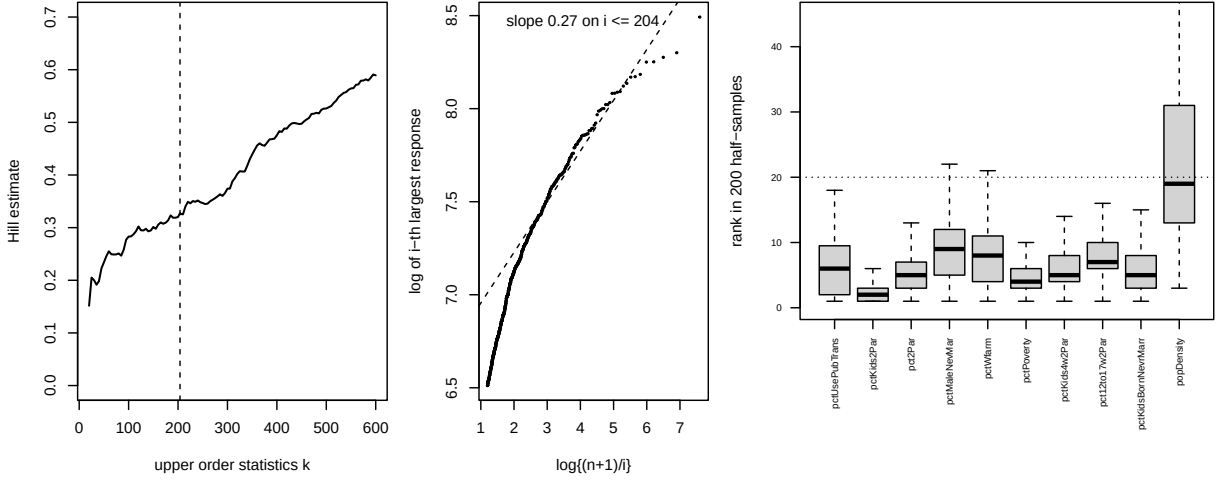


Figure 6: Communities and Crime application ($n = 1,993$, $p = 99$, $\alpha = n^{-0.3}$, $h = n^{-0.1}/2$). Left: Hill estimates for the violent-crime rate; the dashed line marks the working level $k = n\alpha = 204$. Middle: Pareto QQ plot of the 600 largest responses; the dashed line is fitted over the working range $i \leq 204$. Right: distribution of the ranks of the ten leading predictors over 200 seeded half-samples; the dotted line is the default retained size $d = 20$.

related `pct2Par` and `pctKids4w2Par` and the poverty rate `pctPoverty` behave similarly (top-five frequencies .64, .52 and .71). These four variables are also ranked highly by both competing screens (Yoshida–Umezū ranks 3–14, quantile SIS at $\tau = .95$ ranks 1–5), so the three methods agree that the body and the tail of the crime distribution share their main covariates. The remaining primary top-ten variables (`pctMaleNevMar`, `pct12to17w2Par`, `pctKidsBornNevrMarr`) sit in the same demographic cluster but with lower half-sample stability (top-five frequencies .21–.54), and `popDensity`, tenth in the primary fit, is unstable (top-five frequency .04); we group all four with the candidates rather than the findings.

The disagreements are the informative part. The share of workers using public transportation (`pctUsePubTrans`) ranks first in the primary fit and sixth in median over half-samples, yet ranks 57th for the Yoshida–Umezū screen and 53rd for the quantile screen; the farm-employment rate `pctWfarm` shows the same pattern (rank 5 against 60 and 59). These predictors receive relatively strong tail-index screening scores while ranking much lower under the 0.95-quantile screen: their low scores are consistent with marginal variation in upper-tail heaviness across their values, of a kind that a fixed-quantile screen is not designed to detect. Section 4.4 produces the same dissociation under controlled conditions, where the target is known. Both variables are, however, clearly less stable than the family-structure block: their top-five half-sample frequencies are .44 and .35, their 10–90% half-sample rank intervals are $[1, 14]$ and $[2, 19]$, and their worst ranks over the 3×3 tuning grid are 20 and 18, driven by the $a = .25$ fits. We report them as candidate tail-specific predictors for the second-stage analysis to examine.

Ranks are insensitive to tie breaking (maximal displacement over 20 randomized fits: one position for the leaders). We do not interpret any ranking causally: the screen identifies covariates along which the conditional tail index varies marginally, and confounding among the 99 predictors is certain.

6 Discussion

The screen is a first stage. Its intended pipeline reduces p coordinates to a manageable d , after which a joint method — the central tail-index subspace of Gardes and Podgorny [9], a structured tail regression [17, 18], or multivariate local estimation — can resolve interactions and correlated neighbours among the retained variables. The two stages solve different problems: rapid elimination of evidently irrelevant coordinates scales to $p \gg n$, while accurate joint tail inference does not.

Three limitations are intrinsic. First, activity without marginal detectability: when every fibre of an active coordinate meets the global argmax set, the score carries no information about it, at any sample size. Proposition 2.2 characterizes this failure exactly, and it is a property of the target, so enlarging d or n cannot repair it; iterative or grouped extensions would require new identification arguments. Second, mixture bias: when the projected slowly varying factor decays logarithmically, condition (E2) restricts the detectable gaps to $\Delta_{\min} \gg 1/\log n$. Appendix A.6 shows that the severity of this restriction is governed by the conditional mass at the fibre maximum. Third, rare favourable configurations: even with a comfortable population gap, the envelope near its maximum is estimable only from observations whose remaining active coordinates are simultaneously near their maximizing values, and the mass of such configurations can decay exponentially in s . Model M4 in Section 4 shows the resulting cost.

The main open theoretical question is the gap between the primitive model (C1)–(C2) and the estimation conditions (E1)–(E3). Projection creates slow variation that the primitive model does not control, and we do not know useful primitive conditions implying uniform triangular-array moduli for all coordinatewise projections. The score functional also invites variants: a minimum or a low quantile of the profile would emphasize depressions confined to short intervals, at the price of new uniformity arguments for empirical profile extrema.

In practice, tuning matters more than the theory prescribes. The tail fraction trades Hill bias against variance; the bandwidth additionally determines how often near-maximizing configurations of the other active coordinates enter a window. We recommend the workflow of Sections 4 and 5: inspect stability of the leading set over a modest (a, b) grid and under half-sampling, retain a generous d , and reserve interpretation for variables that survive these perturbations. Dependence deserves the same caution: it can help, by concentrating active configurations, or hurt, by promoting inactive neighbours, and the direction is design-specific.

Projection turns a tail-index surface into an upper envelope. Variation of that envelope is a scalable marginal signal for tail-index activity, and its absence, characterized by the argmax projections, tells the analyst exactly what a marginal screen can and cannot see.

A Proofs

Throughout the appendix the dimension is written $p = p_n$, and the auxiliary moduli suppressed in the main text reappear:

$$b_n(a) = \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{s \geq a^{-1}, t > 1} \frac{|\log\{\ell_j(ts, u)/\ell_j(s, u)\}|}{\log t}, \quad \omega_n(r, \tau) = \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{\substack{v \in [0, 1] \\ |v - u| \leq r}} \sup_{w \in [\tau, 1 - \tau]} \left| \log \frac{Q_j(w^{-1}, v)}{Q_j(w^{-1}, u)} \right|,$$

so that (E2) reads $b_n(3\alpha) = o(\Delta_{\min})$ and (E3) reads $\omega_n(C_0 h, \tau_n) = o(\Delta_{\min})$ with $\tau_n = (pn)^{-2}$; note that $\omega_n(\cdot, \tau)$ is nondecreasing in its first argument, so (E3) bounds $\omega_n(r, \tau_n)$ for every $r \leq C_0 h$. Put $\delta_n = (n + 1)^{-1}$,

$$m_n^- = \lfloor \{h + \min(h, \varepsilon)\}(n + 1) \rfloor - 1, \quad K_n = \lfloor \alpha m_n^- \rfloor, \quad S_n = 4n + 1.$$

Once $nh \geq 4$ and $\alpha m_n^- \geq 2$, one has $m_n^- \geq nh/2$ and $K_n \geq \alpha m_n^-/2 \geq nah/4$; the rate conditions (7) force both requirements for n large, which is the only regime where the conversion $K_n \asymp nah$ is used. All conditional distributions are the pointwise compatible versions of Section 2, and constants are uniform over the triangular array. The appendix is organized as follows: preliminary lemmas (A.1); projection (A.2); population results (A.3); the finite-sample uniform profile bound (A.4); quadrature, score concentration and screening, including the proofs of Theorems 3.1 and 3.2 (A.5); model classes satisfying (E1)–(E3) (A.6).

A.1 Preliminary lemmas

The first lemma transfers pointwise perturbations to order statistics, as in Gardes and Podgorny [9, Appendix, Lemma 3].

Lemma A.1 (Sorting is sup-norm nonexpansive). *Let $a_1, \dots, a_m$ and $b_1, \dots, b_m$ satisfy $a_i \leq b_i$ for all i . Then $a_{(i)} \leq b_{(i)}$ for all i . Consequently $|a_i - b_i| \leq \epsilon$ for all i implies $|a_{(i)} - b_{(i)}| \leq \epsilon$ for all i .*

Proof. Fix i . At least $m - i + 1$ indices r satisfy $a_r \geq a_{(i)}$, and for each such r , $b_r \geq a_r \geq a_{(i)}$; hence $b_{(i)} \geq a_{(i)}$. For the second claim apply the first to $(a_i - \epsilon, b_i)$ and $(b_i - \epsilon, a_i)$. $\square$

Lemma A.2 (Chernoff bound for exponential means). *Let L_k be the mean of k i.i.d. standard exponentials. For $0 < t \leq 1$, $\mathbb{P}(|L_k - 1| > t) \leq 2e^{-kt^2/4}$.*

Proof. Markov's inequality applied to $e^{\lambda k L_k}$, optimized at $\lambda = t/(1+t)$, gives $\mathbb{P}(L_k > 1+t) \leq e^{-k\{t - \log(1+t)\}}$. The function $\varphi(t) = t - \log(1+t) - t^2/4$ has $\varphi(0) = 0$ and

$$\varphi'(t) = 1 - \frac{1}{1+t} - \frac{t}{2} = \frac{t(1-t)}{2(1+t)} \geq 0 \quad \text{on } [0, 1],$$

so $t - \log(1+t) \geq t^2/4$ on $(0, 1]$. For the lower tail, $\mathbb{P}(L_k < 1-t) \leq e^{-k\{-t - \log(1-t)\}}$ and $-t - \log(1-t) \geq t^2/2 \geq t^2/4$. $\square$

Lemma A.3 (Threshold order statistic of uniforms). *Let $V_{(1)} < \dots < V_{(m)}$ be the order statistics of m i.i.d. standard uniforms, $0 < \alpha \leq 1/3$ and $k = \lfloor \alpha m \rfloor \geq 1$. Then $\mathbb{P}\{V_{(k+1)} > 3\alpha\} \leq e^{-(k+1)/4}$.*

Proof. If $3\alpha \geq 1$ the probability is zero. Otherwise $\{V_{(k+1)} > 3\alpha\} = \{N \leq k\}$ with $N \sim \text{Bin}(m, 3\alpha)$ and mean $\mu = 3\alpha m \geq 3k$. The multiplicative Chernoff bound $\mathbb{P}\{N \leq (1-\delta)\mu\} \leq e^{-\delta^2\mu/2}$ [3, Section 2.2] with $\delta = 2/3$ gives $\mathbb{P}(N \leq \mu/3) \leq e^{-2\mu/9} \leq e^{-2k/3} \leq e^{-(k+1)/4}$, the last step because $8k \geq 3(k+1)$ for $k \geq 1$. $\square$

A.2 Projection

Let $\mathcal{G} \subseteq \sigma(U)$, $T = \gamma(U)$, and $\xi = \text{ess sup}(T \mid \mathcal{G})$, the smallest $\mathcal{G}$ -measurable random variable dominating T almost surely.

Lemma A.4 (Uniform remainder replacement). *Define $M(y) = \mathbb{E}\{c(U)y^{-1/T} \mid \mathcal{G}\}$. Under (C1)–(C2), almost surely,*

$$\left| \frac{\mathbb{P}(Y > y \mid \mathcal{G})}{M(y)} - 1 \right| \leq \rho(y) := \sup_u \left| \frac{L(y, u)}{c(u)} - 1 \right| \longrightarrow 0. \quad (\text{A.1})$$

Proof. By the tower property, $\mathbb{P}(Y > y \mid \mathcal{G}) = \mathbb{E}\{y^{-1/T}L(y, U) \mid \mathcal{G}\}$. Writing $L(y, U) = c(U)\{1 + r(y, U)\}$ with $|r(y, U)| \leq \rho(y)$,

$$|\mathbb{P}(Y > y \mid \mathcal{G}) - M(y)| = |\mathbb{E}\{c(U)y^{-1/T}r(y, U) \mid \mathcal{G}\}| \leq \rho(y)M(y),$$

and $\rho(y) \rightarrow 0$ is (C2). $\square$

The next lemma is the point at which the argument departs from the proof of Proposition 1 in Gardes and Podgorny [9]: their condition (C.1) assumes the law of $\gamma(X)$ absolutely continuous, which identifies $\{\gamma > t\}$ with $\{\gamma \geq t\}$ in their computation. The tilting argument below needs no such assumption and allows purely atomic conditional laws of T — the relevant case for inactive coordinates and constant-index null models.

Lemma A.5 (Tilted-measure concentration). *For $y > 1$ define the random probability measure*

$$K_y^*(A) = \frac{\mathbb{E}\{c(U)y^{-1/T}\mathbf{1}_{\{T \in A\}} \mid \mathcal{G}\}}{M(y)}, \quad A \in \mathcal{B}([\underline{\gamma}, \Gamma]).$$

On a common probability-one event, $K_y^ \Rightarrow \delta_\xi$ weakly as $y \rightarrow \infty$.*

Proof. Since $K_y^*\{T > \xi + \delta\} = 0$ for every $\delta > 0$ by definition of ξ , it suffices to prove, for fixed $\delta \in (0, \underline{\gamma})$,

$$K_y^*\{T \leq \xi - \delta\} \longrightarrow 0 \quad \text{almost surely.} \quad (\text{A.2})$$

Split by the $\mathcal{G}$ -measurable event $G_\delta = \{\xi \geq \underline{\gamma} + \delta\}$. On G_δ^c , $\xi - \delta < \underline{\gamma} \leq T$, so $\{T \leq \xi - \delta\}$ is empty and (A.2) is trivial. On G_δ , set $r = \xi - \delta$, $s = \xi - \delta/2$ and $\pi = \mathbb{P}(T > s \mid \mathcal{G})$. Then $\pi > 0$ almost surely on G_δ : otherwise $\xi' = \xi - (\delta/2)\mathbf{1}_{G_\delta \cap \{\pi=0\}}$ would be a $\mathcal{G}$ -measurable almost-sure upper bound for T , strictly smaller than ξ with positive probability, contradicting minimality of the conditional essential supremum. Bounding numerator and denominator separately, using only $c_- \leq c \leq c_+$ and monotonicity of $g \mapsto y^{-1/g}$,

$$K_y^*\{T \leq r\} \leq \frac{c_+ y^{-1/r}}{c_- \pi y^{-1/s}} = \frac{c_+}{c_- \pi} y^{-(1/r - 1/s)} \longrightarrow 0.$$

A rational sequence $\delta \downarrow 0$ and a countable intersection give a common probability-one event. $\square$

Proof of Proposition 2.1. For fixed $t > 0$,

$$\frac{M(ty)}{M(y)} = \int_{[\underline{\gamma}, \Gamma]} t^{-1/g} K_y^*(dg) \longrightarrow t^{-1/\xi}$$

almost surely, by Lemma A.5 and boundedness and continuity of $g \mapsto t^{-1/g}$ on $[\underline{\gamma}, \Gamma]$. Lemma A.4 transfers the limit to $\mathbb{P}(Y > ty \mid \mathcal{G})/\mathbb{P}(Y > y \mid \mathcal{G})$. Convergence simultaneously for all $t > 0$ follows from rational t , monotonicity of the ratio in t , and continuity of $t \mapsto t^{-1/\xi}$.

For the fibre identity (3), fix $u \in I_\varepsilon$ and apply the above on the probability space $([0, 1]^{p-1}, K_j(u, \cdot))$ with the trivial σ -field; $\xi_j(u)$ is then the essential supremum of $v \mapsto \gamma(u, v)$ under $K_j(u, \cdot)$. Let $m = \max_v \gamma(u, v)$, attained by compactness and continuity. The essential supremum is at most m . Conversely, for $\eta > 0$ the set $\{v : \gamma(u, v) > m - \eta\}$ is open and nonempty, so has positive $K_j(u, \cdot)$ -probability under (S); the essential supremum is at least $m - \eta$ for every $\eta > 0$. $\square$

A.3 Population results

Lemma A.6 (Continuity of the envelope). *Under (C1), each $\xi_j(u) = \max_v \gamma(u, v)$ is continuous on $[0, 1]$, with modulus bounded by the modulus of continuity of γ .*

Proof. γ is uniformly continuous on the compact cube; if $|u - u'| \leq \delta$ then $|\gamma(u, v) - \gamma(u', v)| \leq \varpi(\delta)$ for every v , and taking maxima preserves the bound. $\square$

Proof of Proposition 2.2. Claim 1. For $j \notin A$, $\gamma(u, v) = g\{(u, v)_A\}$ does not involve u , and as v ranges over the fibre the active subvector ranges over all of $[0, 1]^s$; by (3), $\xi_j(u) = \max_w g(w) = \gamma^*$ for every u .

Claim 2. By (3), $\xi_j(u) = \gamma^*$ exactly when the fibre through u intersects M , i.e. when $u \in \pi_j(M)$.

Claim 3. $u \mapsto \gamma^* - \xi_j(u)$ is continuous (Lemma A.6), nonnegative, and by Claim 2 vanishes on I_ε exactly on $I_\varepsilon \cap \pi_j(M)$. A continuous nonnegative function has positive integral over I_ε if and only if its positivity set has positive Lebesgue measure.

Final statement. Write M^A for the maximizer set of g in $[0, 1]^s$; then $M = \{u : u_A \in M^A\}$ and, for $j \in A$, $\pi_j(M)$ is the projection of M^A , a finite set when M^A is finite; its intersection with I_ε is Lebesgue-null and Claim 3 applies. $\square$

Proof of Proposition 2.3. Every x on the fibre $x_j = u$ has $\|x_A - u_A^*\| \geq |u - u_j^*|$, so every point of the fibre satisfies $\gamma(x) \leq \gamma^* - c|u - u_j^*|^\nu$; taking the fibre maximum gives $\xi_j(u) \leq \gamma^* - c|u - u_j^*|^\nu$. Averaging over $u \in I_\varepsilon$ gives the score bound. For $\nu = 2$, $\mathbb{E}(V - u_j^*)^2 = \text{Var}(V) + (1/2 - u_j^*)^2 = (1 - 2\varepsilon)^2/12 + (1/2 - u_j^*)^2$. $\square$

A.4 Uniform profile bound

Inside proofs the empirical-rank quantities carry a superscript R when the distinction from frozen-location quantities matters. The scheme parallels the proof of Theorem 3 in Gardes and Podgorny [9] — oscillation control, exponential-spacings representation, slowly varying remainder — with the covering net over their compact matrix class replaced by exact union bounds over p coordinates and S_n window states, and their density conditions (H.1)–(H.2) replaced by the deterministic rank geometry.

Lemma A.7 (Rank displacement). *Let $\widehat{U}_{ij} = R_{ij}/(n+1)$ and*

$$D_n(\eta) = \left\{ \max_{j \leq p} \max_{i \leq n} |U_{ij} - \widehat{U}_{ij}| \leq \eta + \delta_n \right\}.$$

Then $\mathbb{P}\{D_n(\eta)^c\} \leq 2pe^{-2n\eta^2}$.

Proof. Fix j and let $F_{n,j}$ be the empirical distribution function of the U_{ij} , $i \leq n$. By the Dvoretzky–Kiefer–Wolfowitz inequality with Massart’s constant [14], $\sup_t |F_{n,j}(t) - t| \leq \eta$ outside probability $2e^{-2n\eta^2}$. On that event, continuity gives $F_{n,j}(U_{(r)j}) = r/n$, hence $|U_{(r)j} - r/n| \leq \eta$ and

$$|U_{(r)j} - r/(n+1)| \leq \eta + \frac{r}{n(n+1)} \leq \eta + \delta_n$$

for every rank r simultaneously. A union bound over j finishes; no union over ranks is needed. $\square$

Lemma A.8 (Deterministic window counts). *For every j , every realization and every $u \in I_\varepsilon$, $M_j(u) \geq m_n^-$ and $k_j(u) \geq K_n$; moreover $M_j(u)$ is a nonrandom function of u .*

Proof. The ranks form a permutation of $1, \dots, n$, so $M_j(u) = \#\{r \in \{1, \dots, n\} : |r/(n+1) - u| \leq h\}$, a deterministic function of u : the number of integers in $[x(u), y(u)]$ with $x(u) = \max\{(u-h)(n+1), 1\}$ and $y(u) = \min\{(u+h)(n+1), n\}$. Fix $u \in I_\varepsilon$ and write $m^- = m_n^- = \lfloor \{h + \min(h, \varepsilon)\}(n+1) \rfloor - 1$; note $h + \min(h, \varepsilon) \leq \min(2h, h + \varepsilon)$. Four cases.

If neither endpoint is clipped, the interval has length $2h(n+1)$ and contains at least $\lfloor 2h(n+1) \rfloor - 1 \geq m^-$ integers.

If only the left endpoint is clipped, then $x(u) = 1$ and the count is $\lfloor y(u) \rfloor$. Either $y(u) = (u+h)(n+1) \geq (\varepsilon+h)(n+1)$, so the count is at least $\lfloor (h+\varepsilon)(n+1) \rfloor \geq m^- + 1$; or $y(u) = n$ and the count is $n \geq \lfloor (h+\varepsilon)(n+1) \rfloor - 1 \geq m^-$, using $h \leq 1 - \varepsilon$, i.e. $(h+\varepsilon)(n+1) \leq n+1$.

If only the right endpoint is clipped, then $y(u) = n$ and the count is $n - \lceil x(u) \rceil + 1 \geq n - x(u) \geq n - (1 - \varepsilon - h)(n+1) = (h+\varepsilon)(n+1) - 1$, hence at least $\lceil (h+\varepsilon)(n+1) \rceil - 1 \geq m^-$.

If both endpoints are clipped, the count is n , which is at least m^- as in the second case. In every case $M_j(u) \geq m^-$, and $k_j(u) = \lfloor \alpha M_j(u) \rfloor \geq \lfloor \alpha m_n^- \rfloor = K_n$. $\square$

Lemma A.9 (Finite-state structure). *For fixed j and fixed ranks, $u \mapsto \mathcal{W}_j(u)$ takes at most $S_n = 4n + 1$ distinct values on $[0, 1]$.*

Proof. The membership indicator of observation i changes only at $\widehat{U}_{ij} \pm h$: at most $2n$ breakpoints, hence at most $2n + 1$ constancy intervals plus the $2n$ breakpoint states. $\square$

Lemma A.10 (Conditional PIT and Rényi representation). *Fix j and condition on the column $\mathbf{U}_j = (U_{1j}, \dots, U_{nj})$. Under (E1):*

1. $V_{ij} = 1 - F_{Y|j}(Y_i | U_{ij})$, $i \leq n$, are i.i.d. standard uniform, and $Y_i = Q_j(1/V_{ij}, U_{ij})$ almost surely;
2. for any index set $\mathcal{W}$ of size m , deterministic given $\mathbf{U}_j$, with $V_{(1)} < \dots < V_{(m)}$ the ordered PIT values of its members and $k = \lfloor \alpha m \rfloor$,

$$L_k = \frac{1}{k} \sum_{r=1}^k \log \frac{V_{(k+1)}}{V_{(r)}}$$

is distributed as the mean of k i.i.d. standard exponentials, independent of $V_{(k+1)}$.

Proof. (i) Given $\mathbf{U}_j$, the pairs (U_{ij}, Y_i) are independent in i and Y_i has continuous conditional distribution function $F_{Y|j}(\cdot | U_{ij})$, so V_{ij} is standard uniform. For the representation, $Q_j(1/V_{ij}, U_{ij}) = F_{Y|j}^{-1}(F_{Y|j}(Y_i | U_{ij}) | U_{ij}) \leq Y_i$ always, with strict inequality only when Y_i lies in the interior of an interval on which $F_{Y|j}(\cdot | U_{ij})$ is constant; each such interval has conditional probability zero and there are countably many, so the representation holds almost surely. Strict monotonicity is not needed. Conditioning on the whole column, rather than on window counts as in Gardes and Podgorny [9, Appendix, Lemma 4], makes the window sets deterministic (Lemma A.8); only coordinate j is conditioned on, and no independence across coordinates is used.

(ii) With $W_i = -\log V_i$ standard exponentials and order statistics $W_{(1)} < \dots < W_{(m)}$, $W_{(m-r+1)} = -\log V_{(r)}$, Abel summation gives

$$kL_k = \sum_{r=1}^k \{W_{(m-r+1)} - W_{(m-k)}\} = \sum_{i=1}^k i \{W_{(m-i+1)} - W_{(m-i)}\},$$

and by the Rényi representation of exponential order statistics [16] the normalized top spacings $i\{W_{(m-i+1)} - W_{(m-i)}\}$, $i \leq k$, are i.i.d. standard exponential and independent of $W_{(m-k)}$, hence of $V_{(k+1)}$. $\square$

Lemma A.11 (Frozen-location Hill expansion). *Fix j , condition on $\mathbf{U}_j$, and fix a window state $\mathcal{W}$ of size m realized at $u \in I_\varepsilon$, with $k = \lfloor \alpha m \rfloor$ and Hill statistic H . On the event $D_n(\eta)$ intersected with $\{V_{ij} \in [\tau, 1 - \tau] \forall i\}$ and $\{V_{(k+1)} \leq 3\alpha\}$,*

$$|H - \xi_j(u)| \leq 2\omega_n(h + \eta + \delta_n, \tau) + \xi_j(u)|L_k - 1| + b_n(3\alpha)L_k. \quad (\text{A.3})$$

Proof. Spatial freezing. Each $i \in \mathcal{W}$ has $|\hat{U}_{ij} - u| \leq h$, hence $|U_{ij} - u| \leq r_n := h + \eta + \delta_n$ on $D_n(\eta)$. With $V_{ij} \in [\tau, 1 - \tau]$, the definition of ω_n gives $|\log Q_j(1/V_{ij}, U_{ij}) - \log Q_j(1/V_{ij}, u)| \leq \omega_n(r_n, \tau)$ for every member. Let $\check{H}$ be the Hill statistic of the frozen responses $\check{Y}_i = Q_j(1/V_{ij}, u)$. By Lemma A.1, ordered log-responses differ by at most $\omega_n(r_n, \tau)$, and the Hill statistic is an average of differences of two of them, so $|H - \check{H}| \leq 2\omega_n(r_n, \tau)$.

Exact decomposition. Since $s \mapsto Q_j(s, u)$ is nondecreasing, $\check{Y}_{(m-r+1)} = Q_j(1/V_{(r)}, u)$, and $\log Q_j(s, u) = \xi_j(u) \log s + \log \ell_j(s, u)$ yields

$$\check{H} = \xi_j(u)L_k + R, \quad R = \frac{1}{k} \sum_{r=1}^k \log \frac{\ell_j(1/V_{(r)}, u)}{\ell_j(1/V_{(k+1)}, u)}.$$

Remainder. On $\{V_{(k+1)} \leq 3\alpha\}$, each ratio is $\ell_j(ts, u)/\ell_j(s, u)$ with $s = 1/V_{(k+1)} \geq (3\alpha)^{-1}$ and $t = V_{(k+1)}/V_{(r)} > 1$, so $|\log(\cdot)| \leq b_n(3\alpha) \log(V_{(k+1)}/V_{(r)})$; averaging gives $|R| \leq b_n(3\alpha)L_k$. $\square$

Theorem A.12 (Finite-sample uniform profile bound). *Assume (C1)–(C2), (S), the structural part of (E1) (continuity of every $F_{Y|j}(\cdot | u)$), hence the quantile representation), that $b_n(3\alpha)$ and $\omega_n(\cdot, \cdot)$ are finite, $0 < h \leq 1 - \varepsilon$, $0 < \alpha \leq 1/3$, and $K_n \geq 2$. For every $\eta > 0$, $0 < t \leq 1$, $0 < \tau < 1/2$,*

$$\begin{aligned} \mathbb{P} \left\{ \max_{j \leq p} \sup_{u \in I_\varepsilon} |\hat{\xi}_j(u) - \xi_j(u)| > 2\omega_n(h + \eta + \delta_n, \tau) + \Gamma t + (1 + t)b_n(3\alpha) \right\} \\ \leq \underbrace{2pe^{-2n\eta^2}}_{\text{ranks}} + \underbrace{2pn\tau}_{\text{PIT truncation}} + \underbrace{2pS_n e^{-K_n t^2/4}}_{\text{spacings}} + \underbrace{pS_n e^{-(K_n+1)/4}}_{\text{threshold}}. \end{aligned} \quad (\text{A.4})$$

Proof. Call a window state *admissible* for coordinate j if it equals $\mathcal{W}_j(u)$ for some $u \in I_\varepsilon$; by Lemmas A.8 and A.9 there are at most S_n admissible states per coordinate and each has $k \geq K_n \geq 2$, so L_k and $V_{(k+1)}$ are well defined on every one of them. (States realized only outside I_ε may have smaller windows; they are simply not needed.) Define the exceptional events $T_1 = D_n(\eta)^c$; $T_2 = \{\exists i, j : V_{ij} \notin [\tau, 1 - \tau]\}$; $T_3 = \{\exists j, \text{ admissible state } \mathcal{W} : |L_k - 1| > t\}$; $T_4 = \{\exists j, \text{ admissible state } \mathcal{W} : V_{(k+1)} > 3\alpha\}$.

Outside $T_1 \cup T_2 \cup T_3 \cup T_4$, fix j and $u \in I_\varepsilon$: the realized window is an admissible state, and Lemma A.11 with $\xi_j \leq \Gamma$, $|L_k - 1| \leq t$, $L_k \leq 1 + t$ gives the displayed deviation bound; the modulus $\omega_n(h + \eta + \delta_n, \tau)$ covers every selected observation because its definition allows the displaced point $v = U_{ij}$ to range over all of $[0, 1]$.

Probabilities: $\mathbb{P}(T_1) \leq 2pe^{-2n\eta^2}$ is Lemma A.7; $\mathbb{P}(T_2) \leq 2pn\tau$ since each V_{ij} is uniform. For T_3, T_4 , condition on $\mathbf{U}_j$; the admissible states are then deterministic with $k \geq K_n$, Lemma A.10 applies to each, and Lemmas A.2 and A.3 give conditional bounds $2e^{-kt^2/4} \leq 2e^{-K_n t^2/4}$ and $e^{-(k+1)/4} \leq e^{-(K_n+1)/4}$, both decreasing in k . A union bound over the at most pS_n admissible states and integration over $\mathbf{U}_j$ complete the proof; overlapping states and dependent coordinates cost nothing beyond the union bound. $\square$

A.5 Scores, concentration, and screening

Recall the score grid $G = \{r\delta_n : r = 1, \dots, n\} \cap I_\varepsilon$, of size $L_n = |G|$, and that $K_n \geq 2$ makes every grid point usable, so $\widehat{\Psi}_j = L_n^{-1} \sum_{u \in G} \widehat{\xi}_j(u)$.

Lemma A.13 (Quadrature). *If $L_n \geq 1$, then for every j ,*

$$\left| L_n^{-1} \sum_{u \in G} \xi_j(u) - \Psi_j \right| \leq q_n := \omega_\xi(\delta_n) + \frac{4\Gamma\delta_n}{1-2\varepsilon}. \quad (\text{A.5})$$

Proof. Write the grid points $u_1 < \dots < u_{L_n}$, consecutive multiples of δ_n . If $L_n = 1$ then $|I_\varepsilon| < 2\delta_n$ and the left side is at most $\Gamma < 4\Gamma\delta_n/|I_\varepsilon|$. Assume $L_n \geq 2$ and partition I_ε exactly: $C_1 = [\varepsilon, u_1 + \delta_n/2]$, $C_{L_n} = [u_{L_n} - \delta_n/2, 1 - \varepsilon]$, and $C_r = [u_r - \delta_n/2, u_r + \delta_n/2]$ otherwise. Interior cells have length δ_n , boundary cells length in $[\delta_n/2, 3\delta_n/2]$, and every point of C_r is within δ_n of u_r . Then

$$\left| \frac{1}{L_n} \sum_r \xi_j(u_r) - \frac{1}{|I_\varepsilon|} \int_{I_\varepsilon} \xi_j \right| \leq \frac{1}{|I_\varepsilon|} \sum_r \int_{C_r} |\xi_j - \xi_j(u_r)| + \Gamma \sum_r \left| \frac{\lambda(C_r)}{|I_\varepsilon|} - \frac{1}{L_n} \right|.$$

The first term is at most $\omega_\xi(\delta_n)$. For the second, both weight vectors sum to one; $|I_\varepsilon| = \sum_r \lambda(C_r) \in [(L_n - 1)\delta_n, (L_n + 1)\delta_n]$ gives $||I_\varepsilon|/L_n - \delta_n| \leq \delta_n/L_n$, so interior cells contribute at most δ_n in total and the two boundary cells at most $\delta_n/2 + \delta_n/L_n \leq \delta_n$ each; the sum is at most $3\delta_n$, and the second term is at most $3\Gamma\delta_n/|I_\varepsilon| \leq 4\Gamma\delta_n/|I_\varepsilon|$. $\square$

Theorem A.14 (Finite-sample score bound). *Under the conditions of Theorem A.12 (with in addition $\omega_\xi(\delta_n) < \infty$) and $L_n \geq 1$, for every $\eta > 0$, $0 < t \leq 1$, $0 < \tau < 1/2$,*

$$\mathbb{P} \left\{ \max_{j \leq p} |\widehat{\Psi}_j - \Psi_j| > B_n(\eta, t, \tau) + q_n \right\} \leq 2pe^{-2n\eta^2} + 2pn\tau + 2pS_n e^{-K_n t^2/4} + pS_n e^{-(K_n+1)/4}, \quad (\text{A.6})$$

where $B_n(\eta, t, \tau) = 2\omega_n(h + \eta + \delta_n, \tau) + \Gamma t + (1+t)b_n(3\alpha)$. If moreover $B_n(\eta, t, \tau) + q_n < \Delta_{\min}/2$, the same right-hand side bounds the probability that some active score fails to precede every inactive score; on the complementary event, $A \subseteq \widehat{A}_d$ for every $d \geq s$, and $\widehat{A}_s = A$.

Proof. Pathwise,

$$|\widehat{\Psi}_j - \Psi_j| \leq \sup_{u \in I_\varepsilon} |\widehat{\xi}_j(u) - \xi_j(u)| + \left| L_n^{-1} \sum_{u \in G} \xi_j(u) - \Psi_j \right|,$$

and Lemma A.13 bounds the second term by q_n on every sample path; reusing the same observations for estimation and averaging therefore costs no extra probability. Theorem A.12 gives (A.6). On the event that every score error is below $\Delta_{\min}/2$: inactive coordinates have $\Psi_j = \gamma^*$ (Proposition 2.2), hence $\widehat{\Psi}_j > \gamma^* - \Delta_{\min}/2$; active coordinates have $\Psi_j \leq \gamma^* - \Delta_{\min}$, hence $\widehat{\Psi}_j < \gamma^* - \Delta_{\min}/2$. Every active score precedes every inactive score, and the screening and recovery statements follow deterministically. $\square$

Proof of Theorem 3.1. Choose $\eta_n = \sqrt{\log(pn)/n}$, $\tau = \tau_n = (pn)^{-2}$, and $t_n = \sqrt{(4/K_n) \log\{pS_n(pn)\}}$. The four probability terms in (A.6) are then each $o(1)$: the first is $2pe^{-2\log(pn)} = 2p(pn)^{-2} \rightarrow 0$; the second is $2pn(pn)^{-2} \rightarrow 0$; the third is $2pS_n\{pS_n(pn)\}^{-1} = 2(pn)^{-1} \rightarrow 0$; the fourth is $pS_n e^{-(K_n+1)/4} \rightarrow 0$ because $\log(pS_n) \leq C \log(pn) = o(K_n)$ by the second condition in (7) and $K_n \geq cn\alpha h$. The same condition makes $t_n \leq 1$ for large n , as required by Lemma A.2, and $\Gamma t_n = O(\sqrt{\log(pn)/(n\alpha h)})$.

It remains to check that the bias part of $B_n(\eta_n, t_n, \tau_n) + q_n$ is $o(\Delta_{\min})$. First, $\log(pn) = o(nh^2)$ gives $\eta_n + \delta_n = o(h)$, so $h + \eta_n + \delta_n \leq C_0 h$ eventually and $2\omega_n(h + \eta_n + \delta_n, \tau_n) \leq 2\omega_n(C_0 h, \tau_n) = o(\Delta_{\min})$

by (E3). Second, $(1+t_n)b_n(3\alpha) \leq 2b_n(3\alpha) = o(\Delta_{\min})$ by (E2). Third, $q_n = \omega_\xi(\delta_n) + O(1/n)$: the first part is $o(\Delta_{\min})$ by (E1), and $1/n \leq \sqrt{\log(pn)/(n\alpha h)}$ since $\alpha h \leq 1 \leq n \log(pn)$, so the second part is absorbed into the stochastic rate. Collecting terms, $\max_j |\hat{\Psi}_j - \Psi_j| \leq O(\sqrt{\log(pn)/(n\alpha h)}) + o(\Delta_{\min})$ outside a vanishing probability. $\square$

Proof of Theorem 3.2 and Corollary 3.3. Under (8), $\sqrt{\log(pn)/(n\alpha h)} = o(\Delta_{\min})$, so the choices in the previous proof give $B_n(\eta_n, t_n, \tau_n) + q_n < \Delta_{\min}/2$ for n large, and the second part of Theorem A.14 applies: the failure probability of active–inactive separation is bounded by the four vanishing terms. Sure screening for every $d \geq s$ and exact recovery at $d = s$ follow. $\square$

A.6 Model classes satisfying the estimation conditions

The two moduli admit a convenient sufficient condition through the logarithmic derivative of the slowly varying factor.

Lemma A.15 (Derivative criterion for the tail modulus). *Suppose that for every j and $u \in I_\varepsilon$ the map $s \mapsto \log \ell_j(s, u)$ is differentiable on $[s_0, \infty)$ with*

$$D(s) := \max_{j \leq p} \sup_{u \in I_\varepsilon} \sup_{s' \geq s} |\partial_{\log s'} \log \ell_j(s', u)| < \infty.$$

Then $b_n(a) \leq D(a^{-1})$ for every $a \leq s_0^{-1}$.

Proof. For $s \geq a^{-1}$ and $t > 1$, the mean value theorem in the variable $\log s$ gives $|\log \ell_j(ts, u) - \log \ell_j(s, u)| \leq D(s) \log t \leq D(a^{-1}) \log t$; divide by $\log t$ and take suprema. $\square$

Proposition A.16 (Finite mixtures with a spectral gap). *Suppose the projected conditional survival functions have the form*

$$\bar{F}_j(y | u) = w_j(u) y^{-1/\xi_j(u)} + \{1 - w_j(u)\} y^{-1/\zeta_j(u)}, \quad y \geq 1, \quad (\text{A.7})$$

with $w_j \in [w_0, 1 - w_0]$, $\gamma \leq \zeta_j < \xi_j \leq \Gamma$, $\inf_{j,u} \{1/\zeta_j(u) - 1/\xi_j(u)\} \geq d_0 > 0$, and w_j, ξ_j, ζ_j uniformly Lipschitz in u . Then (E1) holds, and there are constants depending only on $(w_0, \gamma, \Gamma, d_0)$ and the Lipschitz bounds such that

$$b_n(a) \leq C a^\rho, \quad \omega_n(r, \tau) \leq C r \{1 + \log(1/\tau)\}, \quad \rho = \gamma d_0.$$

Consequently, with $\alpha = n^{-a}$, $h = n^{-b}/2$, $\tau_n = (pn)^{-2}$ and a gap $\Delta_{\min}$ bounded away from zero, (E1)–(E3) and (8) hold for every p with $\log(pn) = o(n^{\min(b, 1-a-b)})$; the n^b branch comes from the spatial modulus (E3), the n^{1-a-b} branch from the local extreme count. For $b > \max(a, 1/3)$ the rank-displacement condition $\log(pn) = o(n^{1-2b})$ of (7) binds before this range, as noted after Corollary 3.3.

Proof. (E1): the survival function (A.7) is continuous and strictly decreasing on $[1, \infty)$, so the conditional cdf is continuous, and the Lipschitz property of ξ_j gives $\omega_\xi(r) \leq Cr$.

For the moduli, write $\bar{F}_j(y | u) = w_j y^{-1/\xi_j} \{1 + g\}$ with $g = g_j(y, u) = \{(1 - w_j)/w_j\} y^{-(1/\zeta_j - 1/\xi_j)}$, so $0 \leq g \leq w_0^{-1} y^{-d_0}$. Solving $\bar{F}_j(y | u) = 1/s$ for the quantile $y = Q_j(s, u)$ and taking logarithms,

$$\log Q_j(s, u) = \xi_j(u) \log\{w_j(u)s\} + \xi_j(u) \log\{1 + g(Q_j, u)\},$$

so that $\ell_j(s, u) = w_j^{\xi_j} \{1 + g(Q_j(s, u), u)\}^{\xi_j}$. A lower bound on the quantile follows from $1/s = \bar{F}_j(Q_j) \geq w_j Q_j^{-1/\xi_j} \geq w_0 Q_j^{-1/\xi_j}$, which gives $Q_j(s, u) \geq (w_0 s)^{\xi_j} \geq c_2 s^\gamma$ for $s \geq w_0^{-2}$, hence $g(Q_j(s, u), u) \leq w_0^{-1} c_2^{-d_0} s^{-\gamma d_0} = C_3 s^{-\rho}$.

Differentiability: $s \mapsto Q_j(s, u)$ is differentiable (inverse of a smooth strictly monotone function), and

$$\partial_{\log s} \log \ell_j(s, u) = \xi_j \frac{g'(Q_j)Q_j}{1+g(Q_j)} \partial_{\log s} \log Q_j = -\xi_j \left(\frac{1}{\zeta_j} - \frac{1}{\xi_j} \right) \frac{g(Q_j)}{1+g(Q_j)} \partial_{\log s} \log Q_j.$$

Since $\partial_{\log s} \log Q_j \in [\zeta_j, \xi_j] \subseteq [\underline{\gamma}, \Gamma]$ (differentiate the identity $\bar{F}_j(Q_j) = 1/s$), $1/\zeta_j - 1/\xi_j \leq 1/\underline{\gamma}$, and $g(Q_j(s, u), u) \leq C_3 s^{-\rho}$, the supremum in Lemma A.15 satisfies $D(a^{-1}) \leq \Gamma \cdot \underline{\gamma}^{-1} \cdot \Gamma C_3 a^\rho = C a^\rho$, whence $b_n(a) \leq C a^\rho$.

Spatial modulus: differentiate the defining identity $\bar{F}_j\{Q_j(s, u) \mid u\} = 1/s$ with respect to u at fixed s :

$$\partial_u \log Q_j(s, u) = - \frac{\partial_u \log \bar{F}_j(y \mid u)|_{y=Q_j}}{\partial_{\log y} \log \bar{F}_j(y \mid u)|_{y=Q_j}}.$$

The denominator lies in $[-1/\underline{\gamma}, -1/\Gamma]$, so its reciprocal is bounded by Γ in absolute value. For the numerator, $\log \bar{F}_j(y \mid u) = \log w_j(u) - \{1/\xi_j(u)\} \log y + \log\{1 + g_j(y, u)\}$, and each u -derivative is bounded using the Lipschitz assumptions: $|\partial_u \log w_j| \leq C$; $|\partial_u \{1/\xi_j\}| \log y \leq C \log y$; and $|\partial_u \log(1 + g_j)| \leq |\partial_u g_j| \leq C g_j(y, u) (1 + \log y) \leq C(1 + \log y)$, since $g_j \leq w_0^{-1}$ and the u -dependence of g_j enters through the Lipschitz functions w_j, ζ_j, ξ_j in its factor and its exponent. At $y = Q_j(w^{-1}, u)$ with $w \geq \tau$, $\log y \leq \Gamma \log(w^{-1}) + C \leq C\{1 + \log(1/\tau)\}$. Integrating $\partial_u \log Q_j$ over $|u - v| \leq r$ gives $\omega_n(r, \tau) \leq Cr\{1 + \log(1/\tau)\}$.

Finally, with $\alpha = n^{-a}$, $h = n^{-b}/2$, $\tau_n = (pn)^{-2}$: $b_n(3\alpha) = O(n^{-a\rho}) \rightarrow 0$; $\omega_n(C_0 h, \tau_n) = O(n^{-b} \log(pn))$, which vanishes exactly when $\log(pn) = o(n^b)$ — this is an additional constraint whenever $a + 2b < 1$, and is implied by $\log(pn) = o(n^{1-a-b})$ when $a + 2b \geq 1$; and $\omega_\xi(1/n) = O(1/n)$. All three are $o(\Delta_{\min})$ for a gap bounded away from zero, and combining with (8) gives the stated range $\log(pn) = o(n^{\min(b, 1-a-b)})$. $\square$

The second class is presented as a derivation outline rather than a theorem: the displayed orders are obtained by the argument sketched below, whose one omitted step is stated explicitly.

Remark A.17 (Plateau-maximizer arrays; derivation outline). Let U have uniform marginals and full fibre support, and suppose that for every j and $u \in I_\varepsilon$ the conditional density of the omitted *active* coordinates given $U_j = u$ is bounded above and below by positive constants on $[0, 1]^{|A \setminus \{j\}|}$; the remaining $p - s$ coordinates may be arbitrarily dependent. For $a_0 \geq 1$, $\kappa > 0$, $\Gamma_0 > \kappa s$, let

$$\gamma(u) = \Gamma_0 - \kappa \sum_{\ell \in A} \psi(u_\ell), \quad \psi(t) = \{\max(t - \tfrac{1}{2}, 0)\}^{a_0},$$

with exact conditional Pareto tails $\mathbb{P}(Y > y \mid U = u) = y^{-1/\gamma(u)}$, $y \geq 1$, so that (C1)–(C2) hold with $L \equiv 1$. Then $\xi_j(u) = \Gamma_0 - \kappa \psi(u)$ for $j \in A$, every active coordinate is detectable, and

$$b_n(\alpha) \leq C \{\log(1/\alpha)\}^{-\{1+(s-1)/a_0\}}, \quad \omega_n(r, \tau) \leq C r \{1 + \log(1/\tau)\},$$

with C depending on the density bounds and degrading geometrically in s . The least favourable case is $s = 2$, where the first exponent is $1 + 1/a_0$; with $\alpha = n^{-a}$, condition (E2) holds whenever $(\log n)^{-\{1+(s-1)/a_0\}} = o(\Delta_{\min})$.

Outline. The fibre maximum is attained on the plateau $\{u_\ell \leq 1/2, \ell \in A \setminus \{j\}\}$, whose conditional probability is bounded below by the density assumption; the conditional law of $\gamma(U)$ given $U_j = u$ is therefore an atom of mass $w_j(u) \geq w_- > 0$ at $\xi_j(u)$ plus a residual whose mass

within δ of the endpoint is of exact order $\delta^{(s-1)/a_0}$, since reaching within δ requires each of the $s-1$ omitted active coordinates to exceed $1/2$ by at most $(\delta/\kappa)^{1/a_0}$. Projecting,

$$\mathbb{P}(Y > y \mid U_j = u) = w_j(u) y^{-1/\xi_j(u)} + \int y^{-1/t} \nu_u(dt),$$

where ν_u is the residual part, whose mass within δ of the endpoint is of order $\delta^{(s-1)/a_0}$. Substituting $1/t - 1/\xi_j \geq (\xi_j - t)/\Gamma_0^2$ and integrating in $x = \log y$ shows the residual integral is $y^{-1/\xi_j(u)} O\{(\log y)^{-(s-1)/a_0}\}$ (a Laplace-type computation with the polynomial mass profile; the omitted step is the uniform matching of upper and lower bounds on ν_u near the endpoint). The quantile inversion and the derivative computation then parallel the proof of Proposition A.16, with $s^{-\rho}$ replaced by $\{\log s\}^{-(s-1)/a_0}$, whose logarithmic derivative gains one further power of $\log s$; Lemma A.15 yields the stated b_n . The omitted step — the uniform two-sided control of the residual mass near the plateau boundary — is elementary but lengthy; the displays above record the resulting orders rather than a complete derivation, which is why this class is stated as a remark.

The continuous-mixture boundary. For $W \sim U(0, 1)$ and $\mathbb{P}(Y > y \mid W) = y^{-1/(1+W)}$, projection gives $Q(s) \sim 4s^2/(\log s)^2$, and the left side of (E2) is of exact order $1/\log(1/\alpha) \asymp 1/\log n$ for $\alpha = n^{-a}$. Comparing the three classes isolates what drives the tail modulus: the conditional mass at the fibre maximum. A spectral gap below the maximum gives a polynomial modulus; an atom with polynomially decaying nearby mass gives $\{\log n\}^{-(1+1/a_0)}$; no atom at all gives $1/\log n$, and gaps of that order or smaller fall outside (E2). This boundary belongs to the problem, not to the proof technique: the projected quantile genuinely converges at a logarithmic rate in the last regime.

References

- [1] Anass Aghbalou, François Portier, Anne Sabourin, and Chen Zhou. Tail inverse regression: Dimension reduction for prediction of extremes. *Bernoulli*, 30(1):503–533, 2024. doi: 10.3150/23-BEJ1606.
- [2] Julyan Arbel, Stéphane Girard, and Hadrien Lorenzo. Shrinkage for extreme partial least-squares, 2024. arXiv:2403.09503v2.
- [3] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, Oxford, 2013. doi: 10.1093/acprof:oso/9780199535255.001.0001.
- [4] Meryem Bousebata, Geoffroy Enjolras, and Stéphane Girard. Extreme partial least-squares. *Journal of Multivariate Analysis*, 194:105101, 2023. doi: 10.1016/j.jmva.2022.105101.
- [5] Abdelaati Daouia, Laurent Gardes, and Stéphane Girard. On kernel smoothing for extremal quantile regression. *Bernoulli*, 19(5B):2557–2589, 2013. doi: 10.3150/12-BEJ466.
- [6] Laurens de Haan and Ana Ferreira. *Extreme Value Theory: An Introduction*. Springer, New York, 2006. doi: 10.1007/0-387-34471-3.
- [7] Jianqing Fan and Jinchi Lv. Sure independence screening for ultrahigh dimensional feature space. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 70(5): 849–911, 2008. doi: 10.1111/j.1467-9868.2008.00674.x.

- [8] Laurent Gardes. Tail dimension reduction for extreme quantile estimation. *Extremes*, 21(1): 57–95, 2018. doi: 10.1007/s10687-017-0300-x.
- [9] Laurent Gardes and Alexandre Podgorny. Dimension reduction for the estimation of the conditional tail index. *Scandinavian Journal of Statistics*, 52(3):1444–1476, 2025. doi: 10.1111/sjos.12792.
- [10] Laurent Gardes and Gilles Stupfler. Estimation of the conditional tail index using a smoothed local Hill estimator. *Extremes*, 17(1):45–75, 2014. doi: 10.1007/s10687-013-0174-5.
- [11] Xuming He, Lan Wang, and Hyokyoung Grace Hong. Quantile-adaptive model-free variable screening for high-dimensional heterogeneous data. *The Annals of Statistics*, 41(1):342–369, 2013. doi: 10.1214/13-AOS1087.
- [12] Bruce M. Hill. A simple general approach to inference about the tail of a distribution. *The Annals of Statistics*, 3(5):1163–1174, 1975. doi: 10.1214/aos/1176343247.
- [13] Runze Li, Wei Zhong, and Liping Zhu. Feature screening via distance correlation learning. *Journal of the American Statistical Association*, 107(499):1129–1139, 2012. doi: 10.1080/01621459.2012.695654.
- [14] Pascal Massart. The tight constant in the Dvoretzky–Kiefer–Wolfowitz inequality. *The Annals of Probability*, 18(3):1269–1283, 1990. doi: 10.1214/aop/1176990746.
- [15] Michael Redmond and Alok Baveja. A data-driven software tool for enabling cooperative information sharing among police departments. *European Journal of Operational Research*, 141(3):660–678, 2002. doi: 10.1016/S0377-2217(01)00264-8.
- [16] Alfréd Rényi. On the theory of order statistics. *Acta Mathematica Academiae Scientiarum Hungaricae*, 4:191–231, 1953. doi: 10.1007/BF02127580.
- [17] Yuya Sasaki, Jing Tao, and Yulong Wang. High-dimensional tail index regression, 2026. arXiv:2403.01318v3.
- [18] Hansheng Wang and Chih-Ling Tsai. Tail index regression. *Journal of the American Statistical Association*, 104(487):1233–1240, 2009. doi: 10.1198/jasa.2009.tm08458.
- [19] Takuma Yoshida and Yuta Umezu. Sure independence screening for covariate-dependent extreme value index estimation, 2026. arXiv:2410.15705v3.